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CRUT
Capital Region Urban Transport

Green Urban Mobility Partnership

COMPREHENSIVE BUS NETWORK OPTIMISATION

CRUT, BHUBANESWAR

odisha

SUM-ACA
Sustainable Urban Mobility
Air Quality, Climate Action & Accessibility

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On behalf of the
German Federal Ministry for Economic Cooperation and Development (BMZ)

COMPREHENSIVE BUS NETWORK OPTIMIZATION

CRUT, BHUBANESWAR

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1. Project Introduction

Rapid urbanization, rising private vehicle ownership, and the spatial expansion of the city have placed increasing pressure on Bhubaneswar's transport systems, making coordinated, efficient, and people-centric mobility planning an urgent priority. In this context, the Bhubaneswar Integrated Mobility Plan (IMP) has been developed as a dynamic and evolving strategy document that consolidates ongoing initiatives, sanctioned projects, and future interventions under a unified mobility vision. The IMP ensures that investments across public transport, active mobility, street design, parking, and governance are complementary, phased, and mutually reinforcing, providing a structured framework for decision-making and institutional coordination. The plan has been prepared under the Sustainable Urban Mobility – Air Quality, Climate Action and Accessibility (SUM-ACA) initiative as part of the Indo-German Green Urban Mobility Partnership.

Within this broader framework, strengthening the efficiency and performance of the city's public transport system has emerged as a critical priority. Bhubaneswar's bus network, operated by Capital Region Urban Transport under the Ama Bus and Ama E-Ride services, has expanded significantly in recent years in terms of route coverage and fleet augmentation, including a gradual shift toward electrification. However, this growth has also introduced operational complexities related to route planning, fleet utilisation, scheduling efficiency, and service reliability. The absence of a data-driven and integrated approach to network planning and operations limits the system's ability to optimise resources, respond to demand variations, and deliver consistent service levels.

In response, a comprehensive Bus Network Optimisation initiative has been undertaken as a priority intervention under the IMP. The scope of this effort includes (i) a detailed assessment of existing routes and operational performance, (ii) development of guidelines for additional fleet deployment to enhance coverage and frequency, and (iii) introduction of software-based solutions for integrated scheduling and operational planning. The objective is to improve service efficiency, reduce operational redundancies, and enable better utilisation of fleet and crew resources while maintaining service reliability.

This initiative represents a critical step toward institutionalising data-driven planning within CRUT's operations. By integrating route rationalisation, fleet deployment strategies, and advanced scheduling tools, the project lays the foundation for a more efficient, reliable, and scalable public transport system. Over time, these improvements are expected to enhance service quality, optimise operational costs, and strengthen the role of public transport as the backbone of Bhubaneswar's mobility ecosystem.

EXISTING ROUTES ASSESSMENT

2. Context

The government of Odisha launched a formal urban public bus service in the year 2011 under the aegis of Bhubaneswar Puri Transport Services Limited (BPTSL). The agency was a subsidiary of Bhubaneswar Municipal Corporation (BMC) and operated buses with a private partner, on the Net-Cost Contract (NCC) model. Though the bus service was very successful initially, owing to the non-viability of the business model and growing costs of operation, the service delivery deteriorated. BPTSL was wound-up as an entity and was rechristened as Capital Region Urban Transport (CRUT) in the year 2018 and operated urban bus services under the name brand of 'MoBus' in Bhubaneswar, Cuttack and Puri.

Learning from failure of the earlier model, CRUT engaged with private operators on the Gross Cost Contracting (GCC) model where the revenue risk is borne by the government agency, while the private operators focus on service delivery and fleet maintenance. CRUT had made great strides in providing affordable and clean bus transport thereby quickly attracting ridership. Understanding the knowledge gaps with respect to bus operations, CRUT had engaged with GIZ as a knowledge partner, to support capacity building of their staff and handholding implementation of various ideas. Within just a year of launching bus operations in 2018, CRUT managed to nearly double its ridership to 1.01 lakh pax/day with 200 buses in operation, translating to an occupancy ratio of about 40% as shown in the graph below.

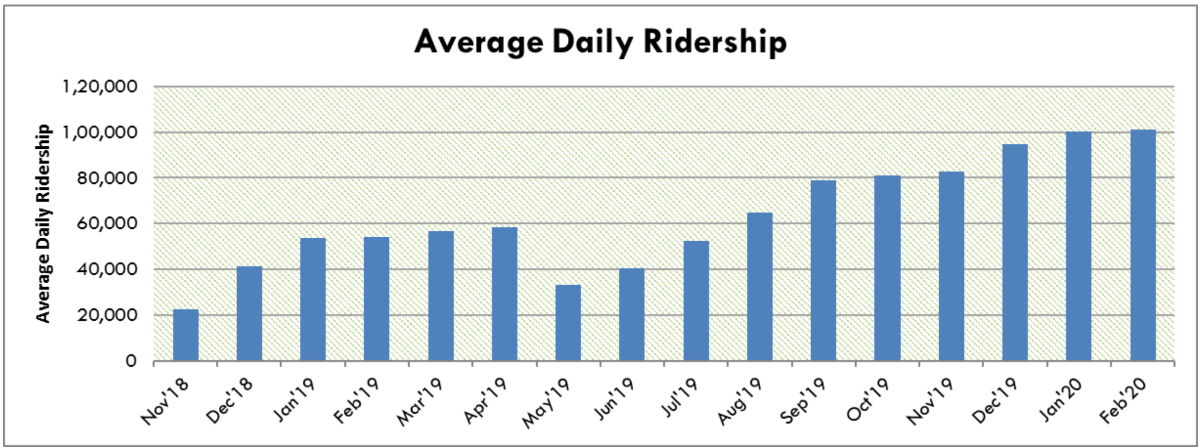


Figure 1: Average daily ridership from Nov'18-Feb'20

Post conclusion of the technical support engagement of GIZ under the SMART-SUT project, and after the withdrawal of the COVID lockdown, CRUT continued to build upon its operational strengths, adding another 100 buses to its fleet and increase in ridership to 2.05 lakh pax/day as of February of 2024. This apart, CRUT has also expanded MoBus operations in Rourkela, Sambalpur and Berhampur in the past year, and is in the process of taking delivery of 200 additional air-conditioned e-Buses for deployment in the Capital Region. In this context, CRUT had once again partnered with GIZ, under the Sustainable Urban Mobility-Air Quality, Climate Action & Accessibility (SUM-ACA) project, to assist in assessing the performance of their current fleet and prepare a deployment for the new bus fleet as well. This report is prepared to address this scope. All the data and analysis presented here is limited to the buses operated from the Capital Region only.

3. CRUT – A Snapshot

Data from the ITS system, primarily the Fare Collection System (FCS) was used for this analysis. Considering various holidays and events happening in the city, **data was collected for the two weeks from 29th of January to 11th of February 2024**, to represent a normal sample. Bus operations related data was collected from the depot level records that were used to make payments to the private operators.

GPS tracking data was also collected for the same period. However, it was found out that the distances reported by the AVLS system in CRUT were erroneous and hence ignored. Further, since the AVLS system only recorded data for every 30 seconds, schedule adherence and related KPIs could not be assessed. Therefore, this data was only used for the bus scheduling exercise (not discussed in this report). Below is a snapshot of the financial/operational performance of MoBus operations in the Capital Region:

Bus Operations Summary:	**Patia**		**Patrapada**	**Pokhriput**	**Chandrasekharpur**	**TOTAL**
Number of Buses	50	50	128	30	50	308
Vehicle Size	12m	12m	9m	9m	9m	
Fuel Type	Diesel	Diesel	Diesel	Diesel	BEV	
Bus Type	AC	Non-AC	Non-AC	Non-AC Refurbished	AC	
Battery Size	-	-	-	-	151 kWh	
Range on Single Charge	-	-	-	-	90-100 km	
Included Spare Buses	3	3	11	3	3	23
Operator	Narbada Travels	Narbada Travels	Travel Time	Travel Time	Jagannathji	
Unit Rate (₹/km)	72.12	57.86	57.13	37.26	70.95	
Assured Distance (km)	175	175	175	175	220	
Rate for Excess Km	0.65	0.65	0.65	0.75	0.5	
Scheduled Utilisation (km)	267	250	222	303	193	236.93
Actual Utilisation (km)	262	249	222	296	176	232.47
Effective Rate (₹/km)	**64.63**	**51.99**	**52.90**	**34.11**	**88.69**	**57.12**

Other Operating Costs:		
Bus Guides @ ₹1k/bus/day (₹/mo):	98,00,000	
Infrastructure Cost (₹/mo):	45,00,000	
Technology Costs (AVLS, AFCS) (₹/mo):	24,00,000	
Sub-Total:	167,00,000	⇒ **₹8.40/km**
Administrative Costs:		
Staff Salaries (₹/mo):	40,00,000	
Communications Cell (₹/mo):	5,00,000	
Administrative Expenditure (₹/mo):	10,00,000	
Sub-Total:	55,00,000	⇒ **₹2.77/km**
TOTAL Other Costs:	**222,00,000**	⇒ **₹11.17/km**

Figure 2: Financial & operational performance of MoBus

Observations:

- Notes:
  - The costs and revenues indicated here are based on most reasonable estimates from the available data. k
  - The infrastructure costs mentioned is a broad estimate calculated assuming depreciation over 20 years.
  - In addition to revenue mentioned here, CRUT receives additional income from interest on deposits and advertising. These incomes are not very significant. At best, it may enable CRUT to offset its losses from operation of the e-Rickshaws.
  - In the absence of reliable disaggregate data, Occupancy Ratio was calculated using total operated kilometres (including dead kilometres). The capacity of 60 pax and 40 pax was assumed for the standard (12m) and midi (9m) buses respectively.
  - Further, the occupancy related calculations do not account for the 12% of passenger trips estimated to be undertaken by 'unlimited travel pass' users.
- CRUT operates **285 schedules daily** along **65 bus routes** and maintains 7.5% of fleet as spare.
- Except for the eBuses, CRUT has attempted to maximise utilisation of buses, achieving an average of **232 km/bus/day**.
- The overall average cost of bus operations is estimated at **₹68.29/km**, of which the ₹57.12 is the average cost paid to the private operator.
- Operating cost of midi eBuses is much more than that of standard size AC diesel buses.
- Further, considering the fleet unitization, **CRUT will need 32% more eBuses to replace the diesel bus fleet**. ⇒ 132 eBuses are required to complete the trips that are being operated by 100 diesel buses.
- In the case of diesel buses, CRUT is able to operate more than the assured kilometres, there by reducing the effective cost per unit distance.

Figure 3: Operations & revenue summary of Mo bus

MoBus Operations & Revenue Summary (Data from 29th Jan to 11th Feb 2024):			
	Non-AC Bus	**AC Bus**	**TOTAL**
Average Daily Ticketed Passengers:	1,26,248	54,454	1,80,702
Average Pass User Trips Per Day (@4 Trips/pass):	16,907	8,129	25,036
Average Daily Ticketed Passenger Kilometers Served (km):	11,21,153	4,69,534	15,90,686
Average Daily Pass Passenger Kilometers Served (km):	1,50,144	70,093	2,20,237
Average Daily Revenue from Ticket Sales:	15,15,733	9,30,212	24,45,944
Average Daily Revenue from Pass Sales:	1,27,724	61,406	1,89,130
Average Ticketed Passenger Trip Length (km):	8.88	8.62	8.80
Average Revenue Per Ticketed Passenger (₹):	12.01	17.08	13.54
Daily Operated Kilometers (km):	45,669	20,586	66,255
Average Service Capacity Supplied (Seat Km):	20,60,820	10,69,720	31,30,540
Average Passengers Per Bus Per Day:	688	626	668
Earning Per Kilometer - EPKM (₹/km):	35.99	48.17	39.77
Occupancy Ratio:	**62%**	**50%**	**58%**

- The non-AC bus operations clock an average OR of 62%, much higher than the 50% observed for the AC buses. This indicates a price-sensitive population. However, it is noteworthy that MoBus occupancy ratio has increased significantly from 40% in 2020.
- In spite of CRUT operating many buses in low-occupancy areas, a **62% average OR** clearly indicates crush loads on many of the routes, especially during peak-hours.
- On an average, MoBus operations are resulting in an **EPKM of ₹39.77/km** from the passenger fare revenue. The reduced EPKM is in part due to 67% of fleet comprising of midi buses.
- Each MoBus caters to **668 pax/bus/day**, an impressive 31% increase over the average of 537 pax/bus/day pre-lockdown (Feb 2020). This is despite all the 108 buses added after 2020 being midi buses.
- The effective average cost of travel by MoBus is ₹1.35/km by non-AC bus and ₹1.98/km by AC bus.
- It can be observed that CRUT is able to **recover about 60% of expenses from fare revenue**. This translates to an estimated annual deficit of ₹68 crore (⇒ ₹ 9.20/pax).
- 12% of MoBus passengers are estimated to use unlimited travel passes offered by CRUT. This includes passengers using physical passes issued at CRUT's office, and those purchased through the mobile app.

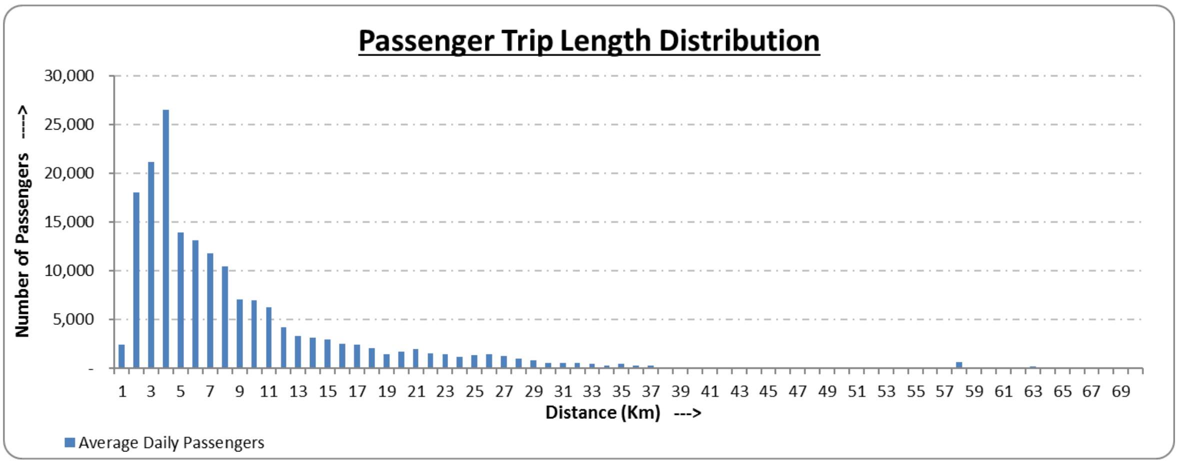
4. Data Trends

Prior to delving deeper into the route level details, this section presents the general tendencies observed from the data that can potentially guide future decision making.

4.1 Travel Distance

As seen in the earlier statistics, the average travel distance of MoBus passengers is 8.80 km, which is relatively high for a city like Bhubaneswar. This anomaly in observation required additional investigation in the form of travel distance distribution. Below is a distribution of passenger trip lengths as extracted from the ticketing data.

Figure 4: Mo bus passengers trip length distribution chart



Passenger Trip Distance Range	% Passengers	% Pax. Kilometers
0 km - 20 km	89.7%	63.4%
20 km - 40 km	8.8%	26.0%
> 40 km	1.5%	10.7%

The visualization above clearly shows that around 50% of all MoBus passengers are travelling between 1 and 6km only and another 20% are traveling between 6 and 10 km. Only 10% of the total passengers travel a distance greater than 20 km and up to 126 km. Thus, the very few long-distance passengers are skewing the analysis. This explains why even though 73% of all the passenger trips are under 10 km length, the average trip length is 8.8 km. It should however be noted that in terms of passenger kilometres, passengers traveling more than 40 km account for 10%. Similarly, those traveling between 20 km and 40 km account for 26% of the total.

Thus, the ticketing data was further stratified and analysed to understand the passenger travel pattern. By separating out the 10% of passenger trips that are over 20 km, The average passenger trip length is found as just 6.29 km and 6.07 km for non-AC and AC buses respectively. It is however possible that some of these passengers are transferring on to another bus to reach their destination. Given most of the ticketing transactions are happening by payment in cash, it is not possible to connect the trips of individuals to understand the actual trip lengths.

This clearly underscores the need for CRUT to incentivise use of cashless means of ticketing such as MoBus Card or Mobile ticketing so that the actual travel pattern can be ascertained.

Figure 5: Cost of travel experienced by passengers

	Non-AC Bus Passenger			AC Bus Passenger		
	Average Trip Length (km)	Revenue Per Ticket (₹)	Revenue Per Pax. Km (₹/km)	Average Trip Length (km)	Revenue Per Ticket (₹)	Revenue Per Pax. Km (₹/km)
Pax. Traveling <= 20km ⇒ 89.7% Tickets	6.29	₹ 9.51	₹ 1.51	6.07	₹ 14.07	₹ 2.32
Pax. Traveling 20 - 40 km ⇒ 8.8% Tickets	28.22	₹ 30.87	₹ 1.09	26.10	₹ 38.41	₹ 1.47
Pax. Traveling > 40km ⇒ 1.5% Tickets	58.78	₹ 55.29	₹ 0.94	60.51	₹ 83.47	₹ 1.38

Another important and interesting aspect to note from the above table is the cost of travel as experienced by the passenger:

- Urban passengers (< 20 km trip length) are paying ₹1.51 and ₹2.32 per kilometre of journey in non-AC and AC buses respectively.
- For the urban passengers, travel by AC bus is 53% more expensive than the non-AC bus travel. This explains the lower occupancy of AC buses.
- In the sub-urban travel range (20 to 40 km), average unit fare of AC bus travel is 35% higher than of non-AC fare. This is commensurate with the ratio of cost of operation of AC and non-AC buses.
- Similarly, the ratio of unit cost for AC vs. non-AC for regional travel (>40 km) is 47%. However, this seems to have arisen due to a much lower than anticipated recovery rate for non-AC bus travel.

4.2 Comparison of Non-AC and AC Bus Ridership

Of the total of 65 bus routes, 21 bus routes have both non-Airconditioned and Airconditioned bus fleet. During the analysis, it has been observed that where both type of services are operated, The AC buses always operate at lower occupancy compared to non-AC buses. The graphs below show the routes in the reducing order of differences in occupancy.

4.2.1 Occupancy Ratio Comparison

It is evident from the data that the difference in the occupancy ratio between non-AC and AC buses varies significantly (25% difference on route #20 and just 1% in route #9) between the routes. One correlation that can be identified is that on routes where the number of AC buses in operation are more than the number of non-AC buses (such as route #9, #11, #23, #12), and where frequency of service is very poor (such as route #51), the difference in occupancy ratio is smaller.

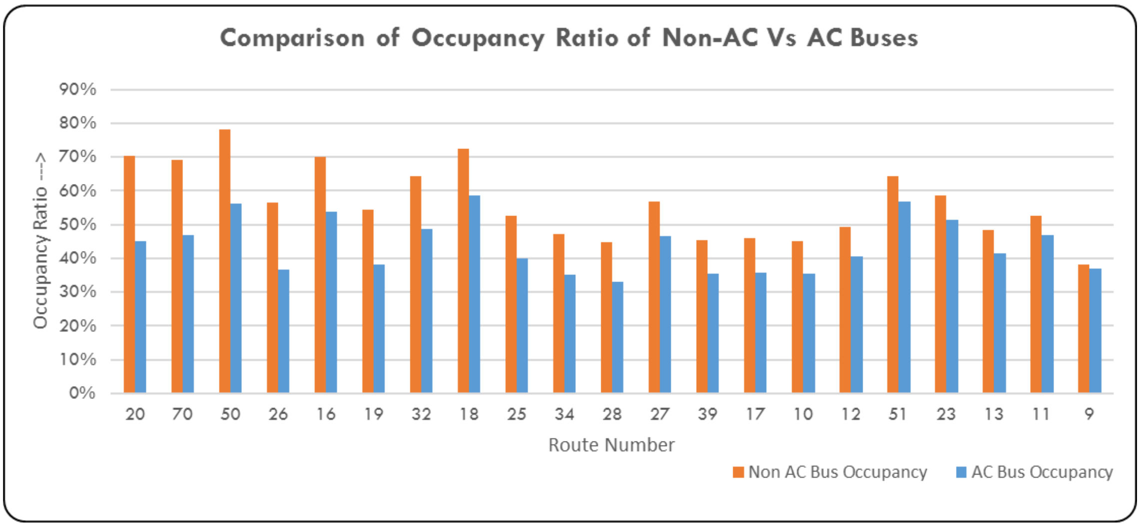


Figure 6: comparison of occupancy ratio in AC & Non-AC buses

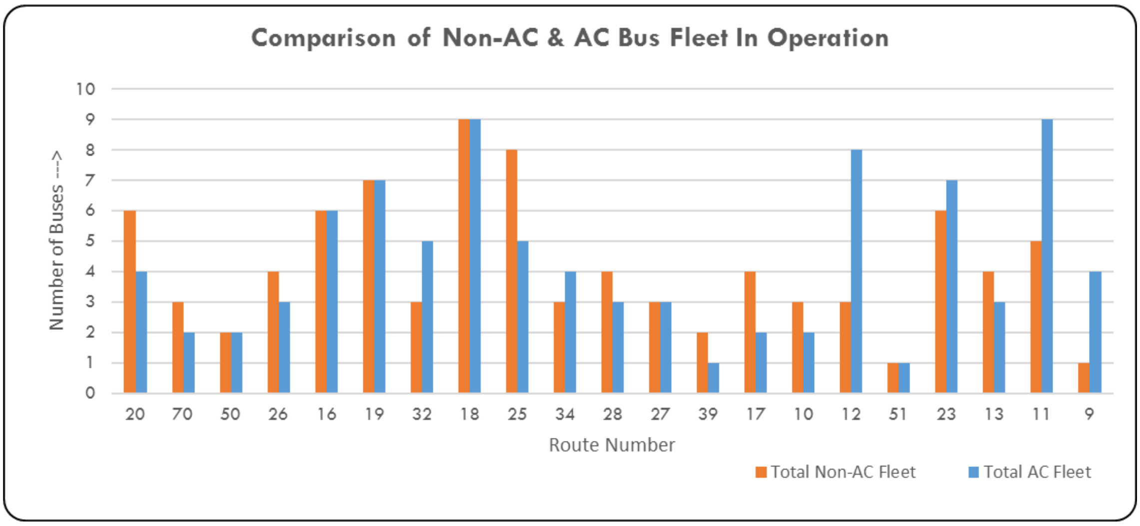


Figure 7: comparison of AC & Non-AC buses in operation

Further, routes #9, #11, and #13 have one end of their route in the contiguous Niladri Vihar/KIIT/Infocity area, thereby having a higher concentration of AC bus fleet in operation in the area, due to which there is nearly no variation in occupancy ration between non-AC and AC buses.

4.2.2 Comparison

Figure 8: Route wise bus fare comparison

Route Length (km)	Non AC Bus Fare	AC Bus Fare	AC Bus Fare Markup
0 - 2	₹ 5	₹ 5	0%
2 - 8	₹5 - ₹10	₹10 - ₹20	50 - 100%
8 - 17	₹15 - ₹20	₹20 - ₹30	33 - 67%
17 - 39	₹25 - ₹40	₹30 - ₹55	20 - 40%
39 - 57	₹45 - ₹55	₹55 - ₹70	18 - 33%
58 - 81	₹60 - ₹75	₹70 - ₹90	13 - 25%
81 - 130	₹80 - ₹120	₹90 - ₹130	8 - 18%

The AC bus travel fares in CRUT are marked higher by an average of 30% in accordance with the higher cost of operation of AC bus fleet. However, as fares are in slabs of ₹5, and in order to maintain a logical structure, the AC bus fares can be double that of the non-AC bus fares in shorter distances and taper down as the travel distance is longer. The table here provides a comparison of the markup. Hence, the expectation is that AC buses also earn about 30% higher fare revenue over the non-AC bus operations. However, a comparison of EPKM indicates otherwise.

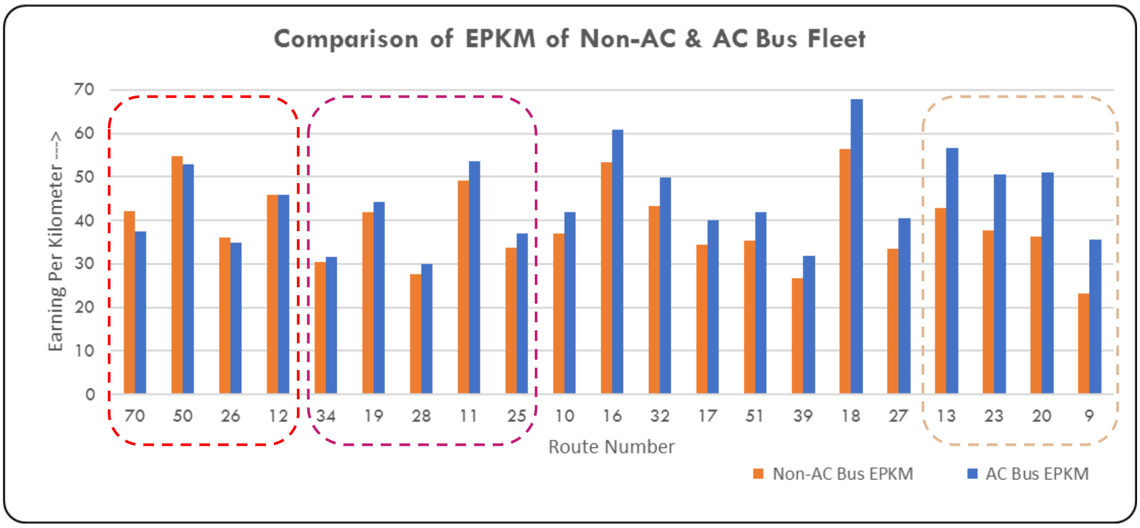


Figure 9: EPKM Comparison chart of MoBus fleet

Of the 21 mixed fleet routes, only 3 routes – route #9, #23, and #13 – are able to generate over 30% premium EPKM on AC bus fleet as compared to the non-AC EPKM. Airconditioned buses on Route #20 also generate 41% higher EPKM than the non-AC fleet. However, that is an anomaly as all the non-AC buses are of midi size (9m) while the AC bus is of standard size (12m).

At the same time, on routes #70, #50, #26, and #12, the EPKM of AC buses is lower than the EPKM of non-AC buses which indicates passenger sensitivity to bus fares in these areas. Similarly, AC buses on routes #34, #19, #28, #11, and #25 also only able to generate 10% premium over the non-AC buses.

Noting this fact, CRUT had significantly reduced AC bus fares during the last revision in December 2020. However, this new evidence indicates that even at current fares, passengers in this region are sensitive to cost of travel.

Given that all the 200 upcoming e-Bus fleet are Airconditioned, CRUT will need a guarded approach to addressing fare sensitivity of passengers in this area.

4.3 Notes for Fare Revision

It should be noted that the AC bus fares were significantly reduced in the previous round of rationalisation in December 2019, which resulted in a much better utilisation and improved revenue from the AC bus fleet. Though the bus fares were not increased since, it is clear that the average fare in AC buses is still not proportional to the cost. **Actions that arise from the above assessment on part of CRUT:**

- CRUT should focus on incentivising switch to cashless transactions, primarily to MoBus Card, in order to better assess passenger journeys, for improved provision of service.
- In order to avoid loss in revenue, the base fares can be increased. However, the discounted fare for the MoBus Card should be planned such that the average fare per unit distance for the non-AC passengers remains at ₹1.50/km, whereas it reduces to about ₹2.10/km for the AC bus travel.
- Upward revision of bus fares should primarily be in the 20+ km travel range, with the target discounted fare with MoBus card for the 20 to 40 km travel is ₹1.15/km and ₹1.60/km for non-AC and AC buses respectively.
- Similarly, for passenger travel greater than 40 km, the discounted unit fare for non-AC and AC buses should be targeted at ₹1.00/km and ₹1.40/km respectively.

4.4 Peak Hours: Temporal Variation in Passenger Demand

Urban passenger demand varies with time with peak demand for public transport typically coinciding with starting and ending time of offices and educational institutions.

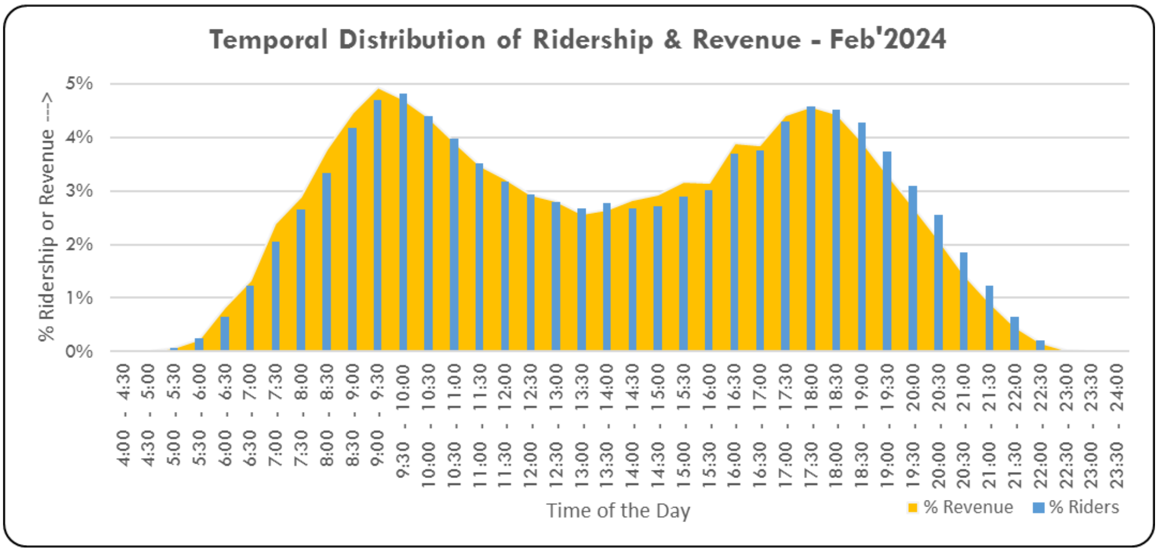


Figure 10: Temporal distribution of Ridership & Revenue

Observations from the consolidated MoBus ticketing data depicted as in above graph reveal two distinct peak periods. The morning peak period extends from 8:00 am to 11:30am, catering to 28.9% of total daily ridership and 29.7% of daily revenue. Similarly, the evening peak hour extends to 3 hours, from 4:00 pm to 7:30 pm. The evening peak period demand stands very close to the morning peak period at 28.8% of daily ridership. **However, the revenue during evening peak period is slightly lesser at 28.5% of daily revenue.**

This is an indication that the average trip length of the evening passengers is slightly smaller than that in the morning. However, this can happen due to various reasons:

- If passenger opt in favour of making transfers between buses in the evening, instead of waiting longer for a direct service.
- If passengers are indeed making shorter trips in the evening, possibly to take care of errands, and then return to their destination by alternative modes.
- The bus guides are unable to promptly issue tickets due to crowding, leading to tickets being issued to the passengers after a few stops have passed.

It is not possible to ascertain the exact reason with the given dataset. If the passengers were instead using digital methods to make payments, then it could have been possible to make the connection, to exactly identify the reason.

More interestingly, the revenue per ticket in the hour preceding the peak periods (7:00 to 8:00 am & 3:00 to 4:00 pm) are higher than the hour after the peak periods (11:30 to 12:30 pm & 7:30 to 8:30 pm). This indicates that people who are making longer journeys tend to start their travel much earlier than those traveling shorter distances.

In order to understand this better, ticketing data has to be grouped into different zones and analysed independently to get more insights.

5. Need for Route Performance Assessment & Review at CRUT

Route performance assessment should indeed be a regular process. However, due to various reasons such as lack of coherent datasets, lack of technical capability, inadequacy of resources to undertake such review, leads to delaying such analysis. Below are some key reasons necessitating this assessment at CRUT now.

- Due to the increasing popularity of MoBus service, CRUT receives many representations from citizens and political representatives to extend services to different parts of the Capital Region and beyond. In the absence of new fleet deployment, CRUT has had to make many ad-hoc changes to routes, spreading the services very sparce. This has sometimes led to deploying single buses along various routes by reducing service in other operational routes.
- It is well known fact that urban bus transport is a loss-making enterprise and requires government funding to be operationally viable. Hence, the financial deficit of CRUT is not entirely unexpected. However, there is a need to ensure that the operations plans (routes, timetables, and bus schedules) are optimal, so as to minimise the deficit, thereby allowing CRUT to use the available funds to add more fleet.
- CRUT has placed an order for 200 additional eBuses which are expected to be delivered in the 2nd half of 2024. At the same time, there are also plans to procure additional e-Rickshaws as well. Hence, this assessment would help in preparing a deployment plan for these new fleet that is arriving.
- The very intent of public transport is to reduce private vehicle dependence of the citizens, thereby reducing traffic congestion and air pollution. However, where a bus is deployed inefficiently, the very purpose of the service is defeated.

6. Route Performance Evaluation – Process & Parameters

There are multiple ways to analyse routes, depending on the depth of correlated data that is available. Given the fragmented way that ITS data is available at CRUT (and almost every other transit agency in India), a reasonably simplistic and repeatable approach has been proposed as follows:

Step - 1: Segregate all the routes into three categories – best performing (top 25%), worst performing (bottom 25%), and the others.

Step - 2: Examine the worst performers in further detail to assess actions to be taken – remove the route, reorganise the route, change timings, or the frequency of service. In cases where the route is being operated due to political or other reasons, the general strategy should be to reduce the vehicle kilometres operated on these routes, unless if some structural changes can result in improvement.

Step - 3: Examine the best performing routes to find opportunities to add more buses (typically those that have been saved from optimising the poorly performing routes). It is important to note that a route that is performing well does not mean that it cannot be further improved.

Performance evaluation of routes is a continuous exercise and should be conducted every quarter. Though the middle 50% of the routes continue to be operated as is, as changes are made to the routes that are worst performing, the focus will come on to a new set of routes that will now form the bottom category in the next quarter.

It is important to note that performance evaluation and bus scheduling together will typically constitute full-time work for one person per 50 routes.

6.1 Key Parameters for Categorising Routes

The choice of parameters for route evaluation depends on the ease of access to disaggregate data. The detail to which secondary data analytics can be performed could be endless. The parameters used for categorising the routes are as follows:

- **Occupancy Ratio (OR):**
  This is the most comprehensive parameter that measures the level to which the capacity of the bus fleet operating on the route is utilised. It takes into account not just the passenger demand but also the extent to which they are traveling along the route, as a ratio of the capacity that is supplied. It is calculated as:

$$\text{Occupancy Ratio} = \frac{Sum\ of\ Individual\ Passenger\ Trip\ Lengths}{Passenger\ Capacity\ of\ Bus\ \ X\ \ Service\ Distance}$$

The passenger capacity of a bus is calculated very differently in different types of services and geographies[1]. In case of a regional/long-distance bus service, the passenger capacity is taken as the number of seats/berths in a bus, but in case of urban travel, traveling by standing is considered acceptable. Typically, in city buses, passenger capacity of a bus is considered as 150% of the number of seats in a bus. However, with new seating configurations coming up, primarily to allow for more standing space in the bus, the same principal may not hold.

For the purposes of CRUT, the capacity of the standard 12-meter-long bus and the 9-meter midi bus should be taken as 60 and 40 passengers respectively. Hence, at a trip level, Occupancy Ratio greater than 1.00 indicates crowding, whereas an OR of 65% means all seats in a bus are occupied.

Calculating the OR however requires storage and access to the disaggregate ticketing data, which may not be available in transit agencies that issue manual tickets, or those who use electronic ticketing machines but only store the summary data.

- **Earning Per Kilometre (EPKM):**
  This is the most commonly available and most basic KPI used by every transit agency and is important in accessing the financial viability gap. Where disaggregate ticketing data is not available, EPKM is used as an proxy for Occupancy Ratio.

$$Earnings\ per\ Kilometre\ (EPKM) = \frac{Total\ Revenue}{Revenue\ Distance + Dead\ Kilometres}$$

  Though intuitively, EPKM may seem proportional to OR, it is not always necessarily so and varies slightly depending on the fare structure followed, and the passenger trip lengths. Hence, it is possible that a route having relatively lower OR still exhibits a relatively higher EPKM.

Since CRUT has a fleet consisting of both AC and non-AC bus fleet operating with different fare structures the benchmarks for EPKM shall also be different. Since the AC bus fares are about 30% higher than the non-AC fares, the threshold EPKM for AC bus fleet will be approximately 30% higher. Similarly, the EPKM is also impacted by the passenger capacity of the bus – thus, the threshold for midi buses is considered 33% lower than the standard bus. Such issues do not however have a bearing when comparing the occupancy ratio. Therefore, the OR is more universal than EPKM.

[1] Tools and Guidelines for City Bus Operations – Part 2 – Planning, Scheduling & Monitoring, Published by GIZ, November 2019

- **Ratio of Route Length to Average Trip Length[2]:**
  This is a very interesting parameter that explains the anomaly in the relationship between OR and EPKM. It is a very useful parameter to quickly assess the type of route, and the remedial actions that may be necessary to improve its performance. The LOS table below explains the nature of the indicator.

Figure 11: LOS table to assess route performances & take remedial actions

	Range	Inference
If the **Ratio of Route Length to Average Trip Length** is	Between 1.0 to 1.2	• This is most symbolic of a long distance or contract bus service • Route should be operated as a non-stop service
	Between 1.2 and 1.8	• Route has potential to be operated as a limited-halts express service
	Between 1.8 and 2.6	• Indicates route functioning optimally for the given demand – continue to monitor changes • Few buses may be operated as limited-halts express service
	Between 2.6 and 3.3	• Further analysis should be done to explore opportunities to create short-turns for better frequency in peak section(s) • Few buses can continue to operate along the whole length of the route
	Above 3.3	• Definite opportunity exists to create short-turns or bifurcation of route ends • Few buses along the route can be divided into two are more parts for operational efficiency

6.2 Challenges in Assessing the Indicators

All the indicators mentioned above are derived from two key datasets – ticketing data, and travel distance data.
While ticketing data is easy to access, it does not capture the travel patterns of the 12% of passengers in CRUT who are traveling using the 'unlimited travel pass'. They are not required to purchase a ticket, and hence that data is missing. Since these users need not be distributed uniformly across the routes, the actual passenger occupancy may be much higher on some of the routes than what is calculated here using the data available. It is hence advisable for bus-transit agencies to discourage use of such travel passes and incentivise passengers to replace them with stored value travel cards, like in the case of

[2] Tools and Guidelines for City Bus Operations – Part 2 – Planning, Scheduling & Monitoring, Published by GIZ, November 2019

metro-rails, where passengers will be required to purchase a ticket even if they are deeply discounted.

Similarly, travel distance data from the ITS system was not very reliable. Hence, distance parameters used are gross values as recorded manually by the depot in-charge. Regular monitoring of technology systems performance and benchmarking can lead to better automated data.

7. Route Performance Analysis

The following sections are entirely derived from the process mentioned above. The ticketing data was processed and summarised route-wise. The detailed spreadsheet with data points segregated by bus type and bus size is provided separately in Annexure-A.

7.1 Identification of Inefficient/Underperforming Routes

As a first step, this section deals with identification of routes that are not most optimal, either in terms of attracting passenger demand, or in terms of the route design itself.

7.1.1 Low Occupancy Routes (Bottom 25%)

The graph below is the list of routes with occupancy ratio less than 40%, an indication of very poor patronage. These routes either require significant alteration or possibly be terminated altogether. In addition to the route, it should be understood that poor patronage could also be a function of the bus timings, frequency of operation, ticket fares, or the local circumstances. Hence, each of these routes need a detailed assessment to come up with suitable action.

It is a common notion among transit agencies to provide buses to every corner of the city even with limited bus fleet, due to 'nobody should be left unserved' approach. However, it should be understood that bus fleet is a valuable commodity and should be deployed to maximise passenger occupancy as that is a clear indicator of the value delivered by the service.

Figure 12: Mobus routes with low Occupancy ratio

					Occupancy Ratio				
Route Number	Route Origin	Route Destination	Route Length	Current TOTAL Fleet	AC/Diesel /Std Bus OR	AC/Ebus/ Midi OR	NAC/Diesel/Std Bus OR	NAC/Diesel/Midi Bus OR	Overall Occupncy Ratio
AE1	Biju Patnaik Internatior	Cuttack Netaji Bus Tern	31.7	1	9%				9%
AE2	Biju Patnaik Internatior	Puri Bus Stand	59.9	1	17%				17%
46	Bhubaneswar Railway S	Nandan Kanan High Sch	25.9	1				18%	18%
45	Bhubaneswar Railway S	Jayadev Pitha Or Kendu	23.0	1				27%	27%
40	Baramunda ISBT	Sbi Colony Kesora	16.4	1				28%	28%
83	Kandarpur	Dhabaleswar	40.3	3				29%	29%
53	Shree Mandira	Maltipatpur	9.7	3				30%	30%
59	Puri Bus Stand	Mahanadi Vihar	89.0	1	30%				30%
36	Bhubaneswar Railway S	Mundali	42.6	1				32%	32%
42	Baramunda ISBT	Nandan Kanan High Sch	23.8	2				32%	32%
61	Baramunda ISBT	Satapada Bus Stand	126.1	1				33%	33%
62	Bhubaneswar Railway S	Gopabandhu Janmapitl	42.7	1				35%	35%
35	Bhubaneswar Railway S	Udayanath College	33.1	3				37%	37%
84	SVNIRTAR Olatpur	Biju Patnaik Park	40.0	1				37%	37%
9	Bhubaneswar Railway S	Patia Square	20.4	5		37%		38%	37%
22B	Khordha New Bus Stanc	Jatani Gate	25.6	3				39%	37%

7.1.2 Low Earning Routes (Bottom 25%)

As indicated earlier, occupancy ratio and earnings need not always be proportional as bus fares are not always proportional to the distance travelled, and there could be multiple fares based on the type of service. Below figure shows the list of routes that are worst performing in terms of earnings. It should be noted here that the selection below takes factors the size and bus fares into consideration.

Figure 13:Low performing routes in terms of EPKM

7.1.3 Disproportionately Long Routes (Worst 25%)

As mentioned earlier, the ratio of the route length (RL) to the average passenger trip length (ATL) is a good indicator of the route design and an easy way to identify routes that can be improved. It has been found that 36 of the 65 routes operated by CRUT have RL/ATL ratio greater than 3.3 indicating many routes that have been extended longer than required based on the passenger travel pattern. In case of CRUT, this has mostly occurred due to original routes being extended far into the outskirts based on political and interest group requests, without examining the demand and reasonableness of such requests. The table below shows 18 routes where the ratio is greater than 4 and has greatest potential to improve the route design. It is worth noting here that some of the routes that appear in this list may have high EPKM (example: Routes – 13, 23, 25, 26, etc). It only means that there is room for further improvement.

Figure 14: Disproportionately long routes in terms of RL/ATL ratio

Route Number	Route Origin	Route Destination	Route Length	Current TOTAL Fleet	Averge Trip Length	RL/ATL
36	Bhubaneswar Railway S	Mundali	42.6	1	7.6	5.6
25	Ranasinghpur	Gadakan Yatra Padia	26.6	13	5.2	5.2
29E	K9B	Sbi Colony Kesora	28.0	0	5.6	5.0
83	Kandarpur	Dhabaleswar	40.3	3	8.2	4.9
47	Settlement Office	IGKC Hospital	39.5	1	8.1	4.9
46	Bhubaneswar Railway S	Nandan Kanan High Sch	25.9	1	5.4	4.8
64	Bhubaneswar Railway S	Jatani Gate	32.0	1	6.6	4.8
24E	Kalinga Vihar	Ranga Bazar Petrol Pum	27.0	0	5.7	4.7
26	Jadupur	Chakeisiani	23.9	7	5.3	4.5
41	Baramunda ISBT	DRIEMS	50.0	2	11.5	4.4
44	Baramunda ISBT	SVNIRTAR Olatpur	42.8	1	9.9	4.3
43	Baramunda ISBT	Banamalipur Medical C	38.7	1	9.0	4.3
80	Agrahat	Police Out Post Naraj	31.2	6	7.4	4.2
23	Bhubaneswar Railway S	IGKC Hospital	21.0	13	5.0	4.2
13	Lingipur	Nandankanan	27.1	7	6.5	4.2
29	Sai Mandir	K9B	22.5	7	5.4	4.2
24	Kalinga Vihar	Sai Mandir	23.2	8	5.6	4.1
38	Bhubaneswar Railway S	Taraboi	43.5	1	10.7	4.1

Route Number	Route Origin	Route Destination	Route Length	Current TOTAL Fleet	Feb'24 Avg EPKM				
					AC/Diesel/Std Bus EPKM	AC/Ebus/Midi EPKM	NAC/Diesel/Std Bus EPKM	NAC/Diesel/Midi Bus EPKM	Overall EPKM from ETM
AE1	Biju Patnaik Internation	Cuttack Netaji Bus Term	31.7	1	14				14
59	Puri Bus Stand	Mahanadi Vihar	89.0	1	20				20
46	Bhubaneswar Railway S	Nandan Kanan High Sch	25.9	1				11	11
60	Bhubaneswar Railway S	Satapada Bus Stand	114.9	1	23				28
61	Baramunda ISBT	Satapada Bus Stand	126.1	1				13	13
AE2	Biju Patnaik Internation	Puri Bus Stand	59.9	1	29				29
83	Kandarpur	Dhabaleswar	40.3	3				16	16
45	Bhubaneswar Railway S	Jayadev Pitha Or Kendu	23.0	1				16	16
62	Bhubaneswar Railway S	Gopabandhu Janmapith	42.7	1				17	17
40	Baramunda ISBT	Sbi Colony Kesora	16.4	1				18	18
36	Bhubaneswar Railway S	Mundali	42.6	1				18	18
48	Jagatpur	Khordha New Bus Stand	57.8	1				18	18
42	Baramunda ISBT	Nandan Kanan High Sch	23.8	2				18	18
53	Shree Mandira	Maltipatpur	9.7	3				18	18
56	Puri Bus Stand	Khordha New Bus Stand	70.1	1				18	18
84	SVNIRTAR Olatpur	Biju Patnaik Park	40.0	1				19	19
35	Bhubaneswar Railway S	Udayanath College	33.1	3				19	19
70	Bhubaneswar Railway S	Konark	62.3	5	37		42		40
58	Puri Bus Stand	Jagatpur	90.0	1				19	19
54	Puri Bus Stand	Biju Patnaik Park	91.2	3				19	19
63	Baramunda ISBT	Madhabananda Temple	56.6	1				20	20
44	Baramunda ISBT	SVNIRTAR Olatpur	42.8	1				20	20

7.1.4 Low Performance Routes

All the routes identified in the above sections offer scope for improvement. However, this does not mean that there is no opportunity for improvement in the remaining routes. The important learning from this exercise is that a route which has high EPKM does not necessarily mean that is already optimally planned or serviced. Suggestions for improvement and quantification of value that can be realised are presented in the section on route-wise recommendations.

7.2 Identification of Best Bus Routes in Capital Region

The best routes identification follows the same process as in case of underperforming routes. In most cases, these routes may need infusion of additional bus fleet to increase frequency of service. This can lead to better service for the passengers and usually should result in increased earnings.

7.2.1 High Occupancy Routes (Top 25%)

The below table lists the top 15 MoBus routes that have an occupancy ratio over 50%, with Route #50 and #82 having the highest occupancy ratio, though primarily for the non-AC bus segment. These routes may potentially cater to more demand with improved fleet and frequencies.

Figure 15: List of high occupancy routes

					Occupancy Ratio				
Route Number	**Route Origin**	**Route Destination**	**Route Length**	**Current TOTAL Fleet**	**AC/Diesel /Std Bus OR**	**AC/Ebus/ Midi OR**	**NAC/Diesel/Std Bus OR**	**NAC/Diesel/Midi Bus OR**	**Overall Occupncy Ratio**
82	Bhubaneswar Railway S	Settlement Office	29.0	5				72%	72%
50	Bhubaneswar Railway S	Puri Bus Stand	57.7	4	56%		78%		67%
18	Baramunda ISBT	Jagatpur	36.6	18	59%		72%		66%
21	Bhubaneswar Railway S	Khordha New Bus Stan	37.4	5				63%	63%
16	Bhubaneswar Railway S	Biju Patnaik Park	34.8	12	54%		70%		62%
51	Baramunda ISBT	Puri Bus Stand	68.4	2	57%		64%		61%
70	Bhubaneswar Railway S	Konark	62.3	5	47%		69%		60%
22A	Bhubaneswar Railway S	Khordha Road Station	22.7	11				61%	59%
20	Bhubaneswar Railway S	Khordha New Bus Stan	35.0	10	45%			70%	58%
23	Bhubaneswar Railway S	IGKC Hospital	21.0	13		51%		59%	55%
81	Jagannath Temple Sale	Barabati Stadium	32.8	7				55%	55%
32	Baramunda ISBT	Lingraj Temple	15.7	8		49%		64%	55%
71	Baramunda ISBT	Konark	73.0	3				55%	55%
27	Bhubaneswar Railway S	AIIMS	14.9	6		46%		57%	52%
30	Bhubaneswar Railway S	Chatabar	27.1	7				51%	51%

7.2.2 High Earning Routes (Top 25%)

As indicated earlier, the EPKM is influenced by the fare structure and the capacity of the bus. Below is the list of 14 highest earning routes in their respective categories.

Figure 16: List of high earning routes

					Feb'24 Avg EPKM				
Route Number	**Route Origin**	**Route Destination**	**Route Length**	**Current TOTAL Fleet**	**AC/Diesel/Std Bus EPKM**	**AC/Ebus/ Midi EPKM**	**NAC/Diesel/Std Bus EPKM**	**NAC/Diesel/Midi Bus EPKM**	**Overall EPKM from ETM**
32	Baramunda ISBT	Lingraj Temple	15.7	8		50		43	47
23	Bhubaneswar Railway S	IGKC Hospital	21.0	13		51		38	44
18	Baramunda ISBT	Jagatpur	36.6	18	68		56		62
82	Bhubaneswar Railway S	Settlement Office	29.0	5				37	37
50	Bhubaneswar Railway S	Puri Bus Stand	57.7	4	53		55		54
20	Bhubaneswar Railway S	Khordha New Bus Stan	35.0	10	51			36	42
26	Jadupur	Chakeisiani	23.9	7		35		36	35
16	Bhubaneswar Railway S	Biju Patnaik Park	34.8	12	61		53		57
11	Bhubaneswar Railway S	Nandan Kanan High Sch	20.8	14	62	45	49		51
12	Bhubaneswar Railway S	Nandankanan	17.1	11	48	44	46		45
22A	Bhubaneswar Railway S	Khordha Road Station	22.7	11				34	34
25	Ranasinghpur	Gadakan Yatra Padia	26.6	13		37		34	35
27	Bhubaneswar Railway S	AIIMS	14.9	6		41		33	37
13	Lingipur	Nandankanan	27.1	7	57		43		49

7.2.3 Routes With Lowest RL/ATL Ratio:

On the routes with lowest RL/ATL ratio, there is not much to do except for operating them as non-stop or limited halts service. However, this may be misleading for express services where tickets are issued end-to-end even if the passenger chooses to alight the bus mid-way. Hence, this analysis is ignored in this case.

7.2.4 Best Performance Routes

The below mentioned 20 routes have been identified as the better performing routes. This does not mean that there is no room for improvement among these routes. Since CRUT operates most routes at very lean frequencies, these routes deserve to be examined to add fleet in general, that will increase the reliability to the passenger, thereby attracting more riders as well as improve the financial viability of MoBus.

11, 12, 13, 16, 18, 20, 21, 23, 25, 26, 27, 30, 32, 50, 51, 70, 71, 81, 82, 22A

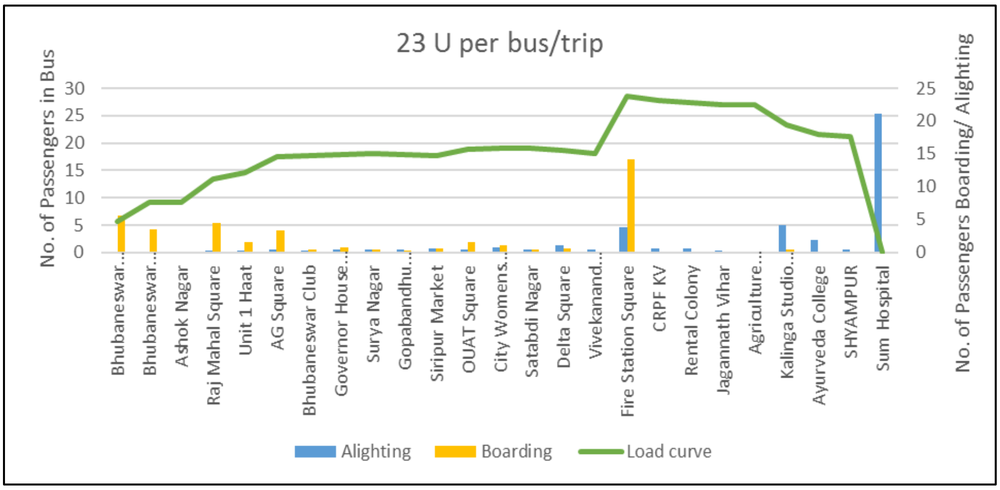
8. Detailed Route Analysis & Recommendations for Revisions

With the routes identified for necessary action, the next step is for a detailed examination to understand the remedial action to be taken to improve the performance. There are two key data visualisations that will help understand the passenger movement pattern along the route.

8.1 Boarding/Alighting & Passenger Load

This visualisation utilises the ticketing data to calculate the average number of passengers boarding and alighting at each bus stop along the bus route on a single trip, thereby allowing to compute the number of passengers onboard along various road segments. Load curve is a visual representation of the number of passengers present in the bus at various points along the route. This visualisation helps in understanding the passenger demand in different sections of the route, thereby allowing to optimise the schedule to match the demand pattern.

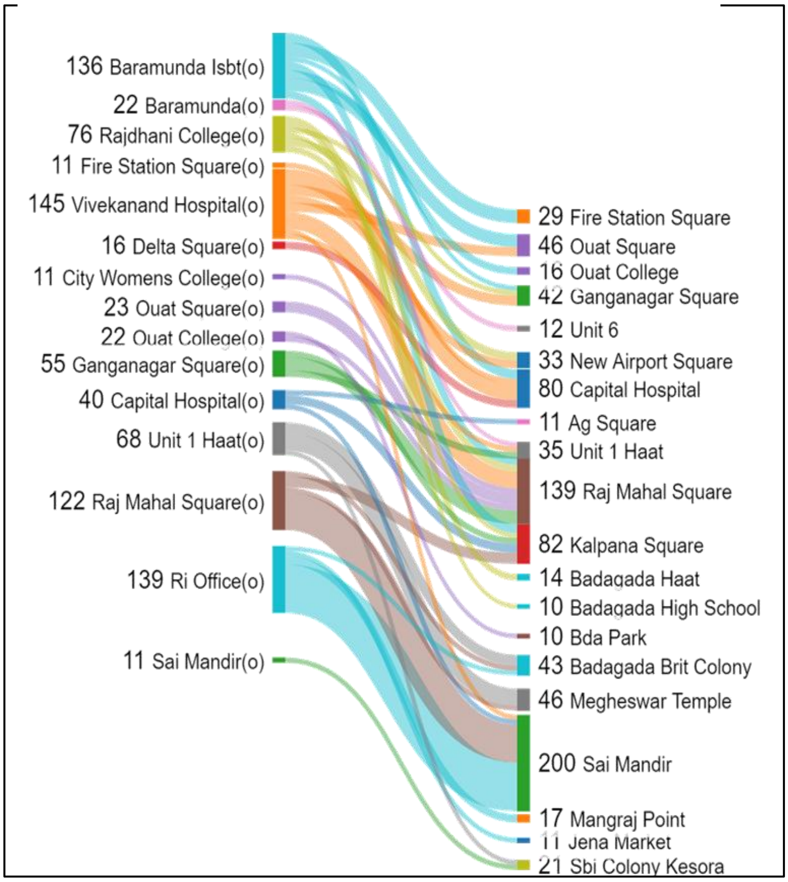
Figure 17: Sample visualisation of route#23 in terms of demand variations



The sample visualisation shown above for route 23 (Up-direction) allows for easy identification of relatively high passenger demand between Fire Station Square and SUM Hospital as compared to the rest of the route.

8.2 Origin-Destination (OD) Visualisation (Sankey Diagram)

Figure 18: Sample OD visualisation of route#40



The passenger load graph visually indicated the number of passengers in the bus in each segment. However, it is not enough to decide the exact segment where to operate extra/fewer trips to minimise passenger transfers. An OD visualisation allows to understand the movement of passengers between the stops on a route.

There are many ways to visualise OD patterns, 'Sankey' diagram being one of them. From the sample visualisation for route 40 (Up) shown here in the figure # 18, it can be seen that almost all the passengers boarding the bus in the first few stops between Baramunda ISBT to Vivekananda Hospital are all alighting the bus before Kalpana Square. This suggests that technically, this route can be operated as two separate overlapping routes – one between Baramunda ISBT to Kalpana Square, and another between Vivekananda Hospital to Sai Mandir. It can also be seen that the very few people travel between Sai Mandir and Kesora, making it not optimal to operate a midi bus at the current frequency.

8.3 Route Geography

As it is evident from the analysis thus far, CRUT has indulged in indiscriminate and ad-hoc extension of bus routes based on requests from various pressure groups. Hence, some of the routes may not follow the most optimal path from passenger travel perspective. Hence, the route map in itself can sometimes explain the anomalies observed in the boarding/alighting, and OD visualisations.

8.4 Route-wise Recommendations

Apart from the three KPIs, the above visual analysis is conducted for each of the routes to come up with the following recommendations. In order to limit the bulk, the visualisations are presented only in a few instances. The first set addresses the low-performing routes, then followed by relatively high performing routes. In both the sections, routes are sequenced in the order of priority by impact.

8.4.1 Low Performance Routes

Route #AE1:

Route: Biju Patnaik Intl Arpt - Cuttack Netaji Bus Term.
Operating Parameters: 31.7 km | 1 Std Bus | AC – Express | 7 One-way Trips
Performance Indicators: **OR:** 9% | **EPKM:** ₹14/km | **Ridership:** 37 pax/bus/day
Assessment: This route does not serve its intended purpose of direct and fast access to airport for multiple reasons:

- Just 3-4 trips in a day is not a dependable service for a passenger traveling to the airport
- This bus is stationed quite far from the arrival gate of the airport, making it difficult for passengers exiting the airport to know that it exists, or its timings, route, or fare.
- Extremely inefficient use of the bus by CRUT at a time that it is facing intense pressure due to lack of fleet.

Recommendation: There are two different approaches by which this bus can be better utilised, depending on CRUT's perspective:

- Scrap the service and reuse it in one of the high-demand, high revenue routes (Ex: Route #18 or #16 – both having occupancy over 50% for AC bus and EPKM over ₹60/km)
- CRUT also operates routes #10 and #17 from the airport which carry more than 600 pax/day and earn over ₹40/km

Impact Estimate: **Ridership Increase:** 600 pax/day
Revenue Increase: ₹2.15 - ₹3.22 lakh/month

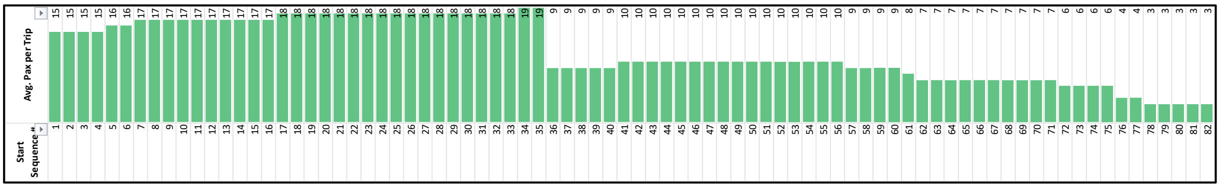
Route #AE2:

Route: Biju Patnaik Intl Arpt - Puri Bus Stand
Operating Parameters: 59.9 km | 1 Std Bus | AC – Express | 6 One-way Trips
Performance Indicators: **OR:** 17% | **EPKM:** ₹29/km | **Ridership:** 66 pax/bus/day
Assessment: Same as route #AE1
Recommendation: Again, two different options for improvement:

- Scrap the service from airport and instead operate it along route #50 starts from Bhubaneswar Rly. Stn. and operates to Puri.
- Alternatively, if CRUT wishes to retain the bus service from the Airport, it is advised to route the bus via Bhubaneswar Rly. Stn. and follow the stops and fares of route #50 instead of operating it as a non-stop express service.

Impact Estimate: **Ridership Increase:** 300 pax/day
Revenue Increase: ₹1.90 lakh/month

Route: Bhubaneswar Rly. Stn. –
114 km | 1 Std Bus| AC |
Performance Indicators: **OR:** 41% | **EPKM:** ₹28/km | **RL/ATL:** 2.2



Ridership: 233 pax/bus/day

Figure 19: Passenger Load – Route #60 (Up): Bhubaneswar Rly. Stn. – Satapada Bus

Assessment: There are multiple challenges with the planning of this route. Firstly, CRUT should not be operating regional routes such as the one to Satapada. Further, the demand between Puri and Satapada is very poor for a one-of service, thus significantly reducing the performance of this route. Route #50 which operates only between Bhubaneswar and Puri on the other hand generates a revenue of ₹55/km.

Recommendation: This route should be terminated and the bus should be added to Route #50, thus bettering the frequency along that route.

Impact Estimate: **Ridership Increase:** 200 pax/day
Revenue Increase: ₹2 lakh/month

ROUTE #61:

Route: Baramunda ISBT – Satapada Bus Stand
Operating Parameters: 126 km | 1 Midi Bus | Non-AC | 4 One-way Trips
Performance Indicators: **OR:** 33% | **EPKM:** ₹13/km | **RL/ATL:** 4.0
Ridership: 193 pax/bus/day

Assessment: Same as Route #60

Recommendation: This route should be terminated and the bus should be added to Route #51 which ₹35/km (for standard bus), thus bettering the frequency along that route. However, it is also recommended that both Route #51 and #61 are routed via Bhubaneswar Rly. Stn., which can further increase revenue up to ₹30/km even in case of a midi-bus.

Impact Estimate: **Ridership Increase:** 130 pax/day
Revenue Increase: ₹2 lakh/month

ROUTE #71:

Route: Baramunda ISBT – Konark
Operating Parameters: 73 km | 3 Midi Bus | Non-AC | 16 One-way Trips
Performance Indicators: **OR:** 55% | **EPKM:** ₹22/km | **RL/ATL:** 2.9
Ridership: 356 pax/bus/day

Assessment: This route is essentially same as Route #70 but starts from Baramunda ISBT instead of Bhubaneswar Rly. but performs much poorly. The Occupancy Ratio stands just at 55% even though it is being operated by midi buses, as compared to 69% for standard non-AC bus in case of route #70. For long distance routes, the aim should be to operate buses at 85% occupancy with a very low RL/ATL ratio. However, route #71 has an RL/ATL value of 2.9 as against 2.3 in case of route #70, which is because the buses on this route operate with fewer passengers until the bus reaches Rasulgarh Square and sees good demand only after Kalpana Square where it merges with Route #70.

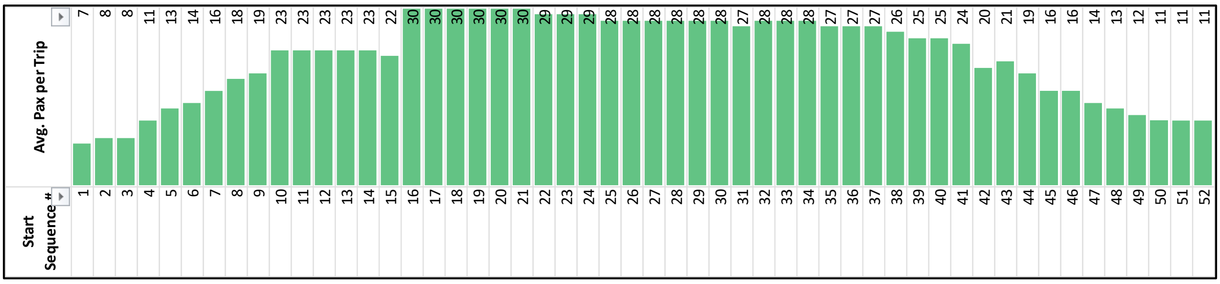


Figure 20: Passenger Load – Route #71 (Up): Baramunda ISBT – Konark Stand Stand

Recommendation: The performance of this route can be significantly improved by routing the route via Bhubaneswar Rly Stn, thereby allowing all the 8 buses (route #70 and #71 combined) to operate in unison, thereby providing enhanced passenger experience.

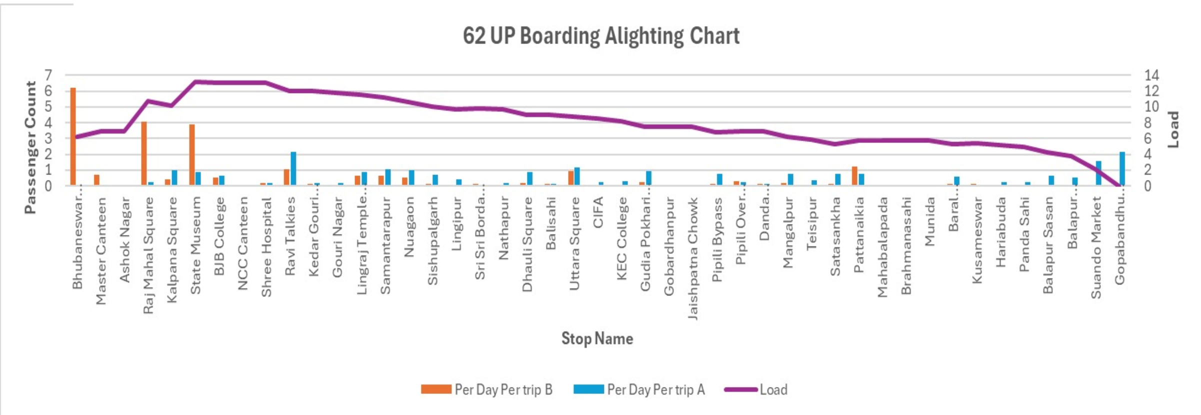
Impact Estimate: **Ridership Increase:** 400 pax/day (3 midi buses together)
Revenue Increase: ₹2.7 lakh/month

ROUTE #62:

Route: Bhubaneswar Rly. Stn. – Gopabandhu Janmapitha
Operating Parameters: 42.7 km | 1 Midi Bus | Non-AC | 6 One-way Trips
Performance Indicators: **OR:** 35% | **EPKM:** ₹17/km | **RL/ATL:** 3.5
Ridership: 333 pax/bus/day

Assessment: This is another classic case of CRUT attempting to server every single destination, which is suitable for an IPT service but not for bus transport. This route follows that of Route #50 until Pattanaikia (36.3 km) but then detours for 6.4 km connect Gopabandhu Janmapitha. But given it services just 3 round trips, this bus is not a practical service for passengers from the extension, nor is it useful for people traveling to Puri. Less than 40 people travel along this route in a whole day.

Figure 21: Boarding & Alighting chart of route#62UP



Recommendation: Terminate this route entirely and CRUT should operate a very effective and frequent trunk service along Bhubaneswar – Puri stretch. This will automatically align the local auto-rickshaws and mini-bus operators to operate a feeder service between Pattanaikia and Gopabandhu Janmapitha, making it a win-win for all stakeholders.

Impact Estimate: **Ridership Increase:** No impact on ridership
Cost Saving: ₹4 lakh/month

ROUTE #54, #58, #59:

Route: Puri Bus Stand - Biju Patnaik Park/Jagatpur/Mahanadi Vihar

Operating Parameters: 89-91 km | 4 Midi Non-AC & 1 Std. AC | 30 One-way Trips

Performance Indicators: **OR:** 30-50% | **EPKM:** ₹19-20/km | **RL/ATL:** 3.4 – 4.0
Ridership: 320 pax/bus/day

Figure 22: Passenger Load – Route #54 (Dn): Puri Bus Stand – Biju Patnaik Park

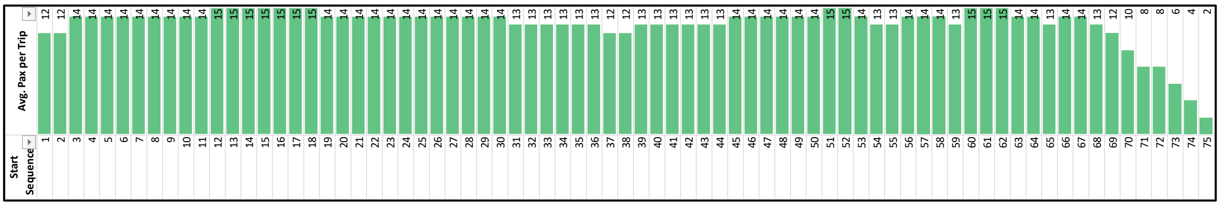


Figure 23: Passenger Load – Route #54 (Up): Biju Patnaik Park – Puri Bus Stand

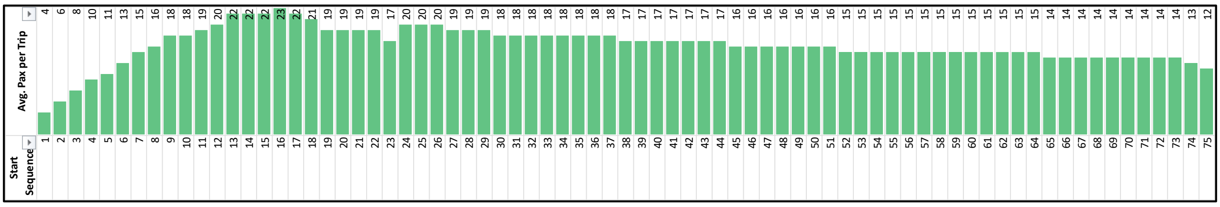
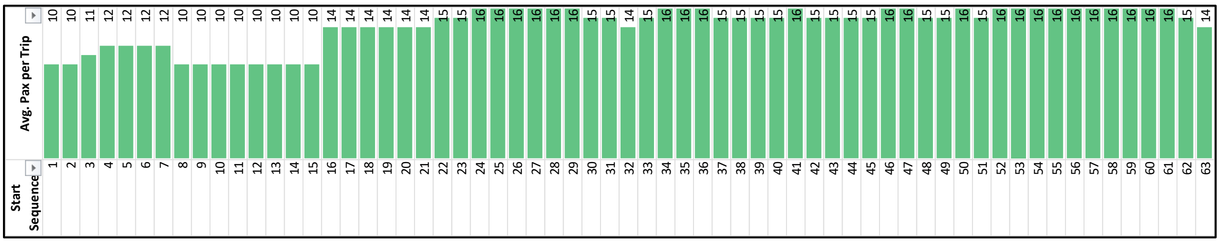


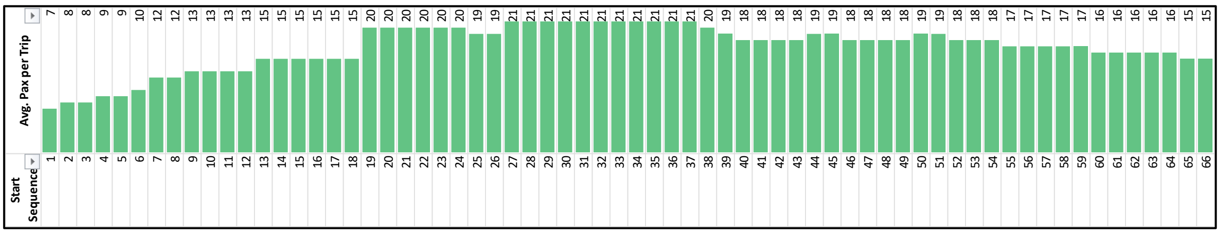
Figure 24: Passenger Load – Route #58 (Up): Jagatpur – Puri Bus Stand



Assessment: In planning these 3 routes, CRUT has erred in terms of combining a predominantly regional route with limited frequency with an urban route section instead of terminating them at the Badambadi Bus Stand – a passenger terminus for long-distance routes. This approach has unnecessarily complicated the route timetabling while at the same time making it difficult for communicating to the passenger and maintaining uniform headways.

Further, route #59 is being operated by an standard size AC bus which does not seem to be finding favour with the passengers on this route.

Figure 25: Passenger Load – Route #59 (Up): Mahanadi Vihar – Puri Bus Stand



Recommendation: Following changes can make it easy for all stakeholders – planners, operators, and passengers:

- Replace the standard AC bus with non-AC midi bus (reduces cost; improves occupancy)
- Operate all 5 buses between Puri Bus Stand and Badambadi Bus Stand (82.2 km) at uniform 60-minute headway (16 trips in each direction starting at 5 am at Cuttack and 7:00 am from Puri). Such ease of communication will attract many passengers. Once the occupancy ratio increases above 70%, the midi buses can be replaced with standard buses.
- The 5 buses are currently operating from Patrapada and Patia depots, which results in significant dead mileage. This route should be operated with 4 buses stationed at Cuttack and 1 bus at Puri depots.

Impact Estimate: **Ridership Increase:** about 900 pax/day (5 midi buses together)
Revenue Increase: ₹6.0 lakh/month

ROUTE #56:

Route: *Puri Bus Stand - Khordha New Bus Stand*
Operating Parameters: *70.1 km | 1 Midi Bus | Non-AC | 6 One-way Trips*
Performance Indicators: ***OR:*** *47% |* ***EPKM:*** *₹18/km |* ***RL/ATL:*** *2.7*
Ridership: *240 pax/bus/day*

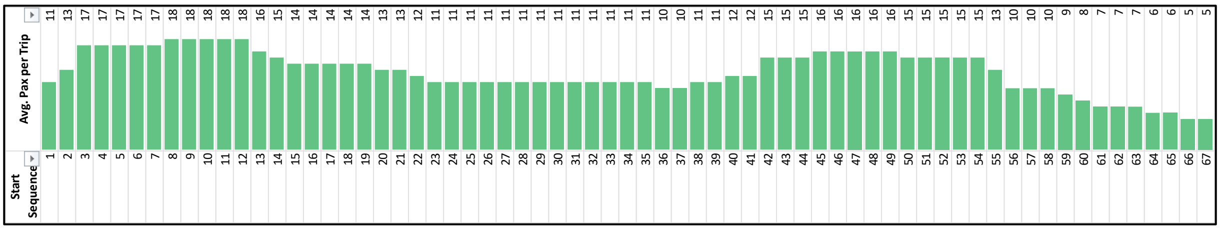


Figure 26: Passenger Load – Route #56 (Dn): Puri Bus Stand – Khordha New Bus

Assessment: This route is essentially 2 separate routes. It can be seen that most people taking this bus at Puri are alighting the bus before Pipli where the bus turns towards Khorda instead of Bhubaneswar.

Recommendation: This route should be terminated and the bus should be operated along route #50 between Bhubaneswar Rly. Stn. and Puri.

Impact Estimate: **Ridership Increase:** No Change (however, pax trip length will double)
Revenue Increase: ₹1.40 lakh/month

ROUTE #57:

Route: Puri Bus Stand - Old Bus Stand Kakatpur

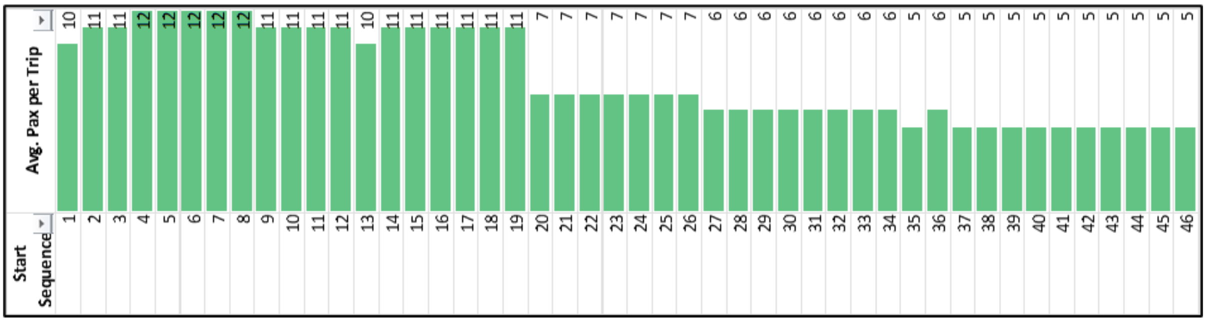
Operating Parameters: 57.6 km | 1 Midi Bus | Non-AC | 6 One-way Trips

Performance Indicators: **OR:** 50% | **EPKM:** ₹21/km | **RL/ATL:** 1.9

Ridership: 235 pax/bus/day

Assessment: The poor performance is once again due to the obvious reason mentioned earlier. Single bus operation with infrequent service.

Figure 27: Passenger Load – Route #57 (Up): Puri Bus Stand - Old Bus Stand Kakatpur



Recommendation: This route should be terminated or else operate only between Puri and Konark (36 km) with two buses operating at a headway of 1 hour or lesser.

ROUTE #46:

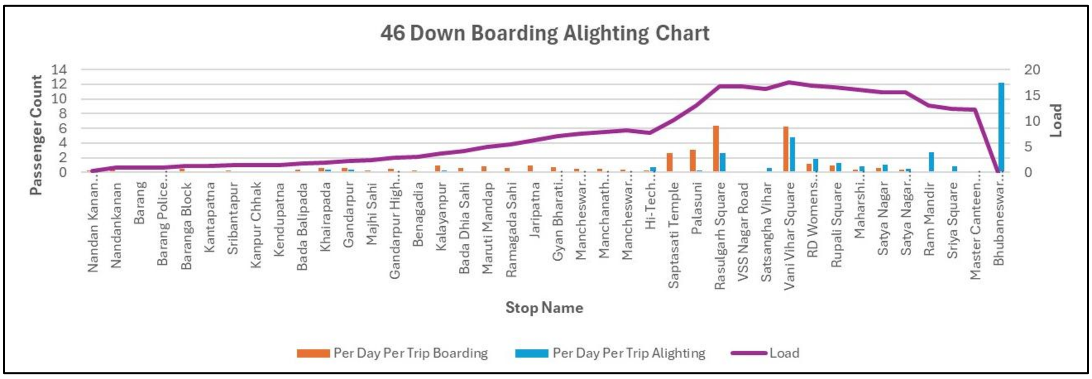
Route: *Bhubaneswar Rly. Stn. - Nandan Kanan High School*

Operating Parameters: *25.9 km | 1 Midi Bus | Non-AC | 8 One-way Trips*

Performance Indicators: ***OR:*** *18% |* ***EPKM:*** *₹11/km |* ***RL/ATL:*** *4.8*

Ridership: *253 pax/bus/day*

Figure 28: Boarding & Alighting chart for route# 46Dn



Assessment: This route seems a clear case of ad-hoc decision making due to interest group pressures.

- As can be seen from the load curve, this bus is virtually unused for the first 18 km between Nandan Kanan High School and Hi-Tech Hospital Square. The average load is under 6 passengers which is better served by an auto-rickshaw rather than a bus.
- It is possible that this route has high vehicular traffic. It needs to be understood that urban travel is usually very short and for a passenger to depend on a bus and wait

for it requires buses operated at very frequent intervals. This route is completely contrary to that understanding and hence ignored by passengers. The route between Rasulgarh Square and Bhubaneswar Rly. Stn. experiences much better ridership as that segment is serviced by trips from many other bus routes.

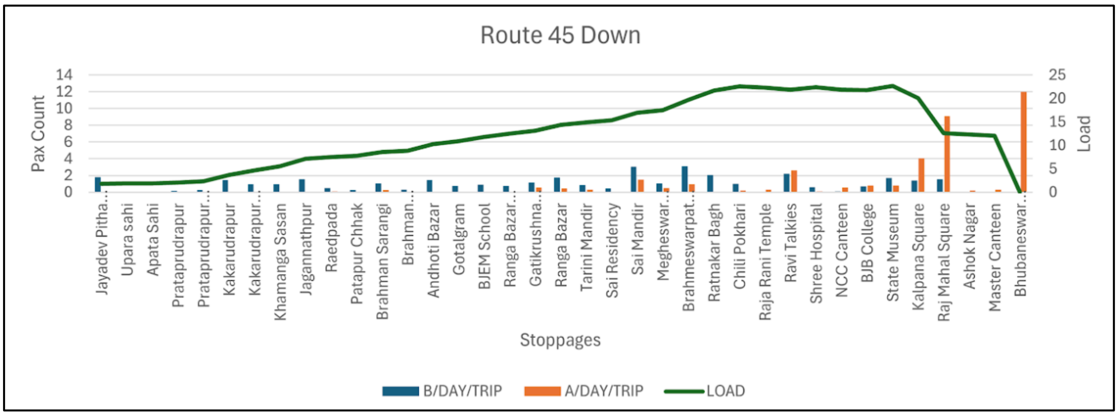
Recommendation: Terminate the route entirely. The bus can be used on alternative routes with high demand (Routes #32, #23, #82, etc.)

Impact Estimate: **Ridership Reduction:** up to 50 pax/day
Fleet Saved: 1 Midi Bus

ROUTE #45:

Route: Bhubaneswar Rly. Stn. - Jayadev Pitha Or Kendubilwa
Operating Parameters: 23.0 km | 1 Midi Bus | Non-AC | 8 One-way Trips
Performance Indicators: **OR:** 27% | **EPKM:** ₹16/km | **RL/ATL:** 3.7
Ridership: 302 pax/bus/day
Assessment: Same scenario as route #46.

Figure 29: Boarding & Alighting pattern at route# 45Dn



Recommendation: There are two possible approaches:

- This route only deserves to be operated between Bhubaneswar Rly. Stn. and Sai Mandir, in which case the route length will reduce to 5.3 km and thus operate 36 one-way trips each day. However, as mentioned earlier, operating a route with a single bus does not work for urban travel patterns and hence not preferred.
- The bus can be merged with existing routes operating to Sai Mandir – Route #24 or #29 or #40 which have EPKM of ₹31/km, ₹26/km and ₹18/km respectively – the preferred option. However, it should be noted that all three routes also have potential to be optimised and will be discussed further.

Impact Estimate: **Ridership Reduction:** up to 50 pax/day
Fleet Saved: 1 Midi Bus

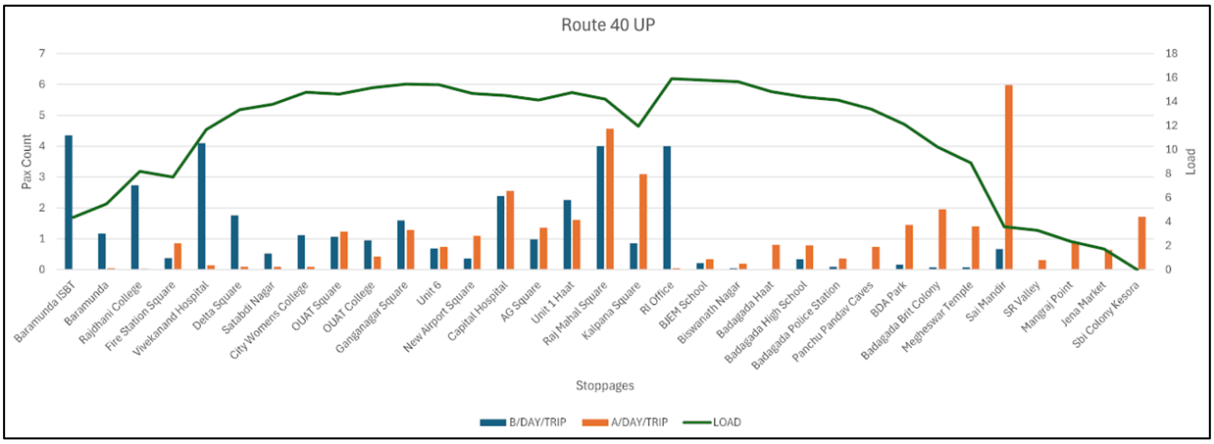
ROUTE #40:

Route: Baramunda ISBT - Sbi Colony Kesora
Operating Parameters: 16.4 km | 1 Midi Bus | Non-AC | 12 One-way Trips
Performance Indicators: **OR:** 28% | **EPKM:** ₹18/km | **RL/ATL:** 3.6
Ridership: 441 pax/bus/day

Assessment: While this route also has lower OR and EPKM, the average daily ridership per bus is closer to the overall average of midi-buses, which indicates a potential to improve if operated with better frequency and in a more optimised way. Only the route segment between Sai Mandir and Kesora – a route length of 5.5 km is underperforming.

Recommendation: Following actions can significantly improve the performance of this route:

Figure 30: Boarding & Alighting pattern on route# 40UP



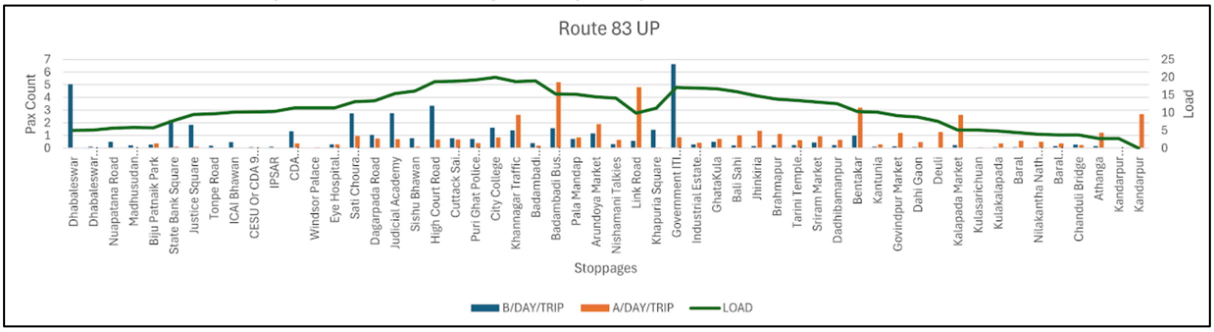
- Operate the route only between Baramunda ISBT and Sai Mandir – 10.9 km. This will also allow a single bus to operate 18 one-way trips instead of the current 12/day. This will bring down the RL/ATL ratio to under 2.4 which will significantly improve occupancy and thus revenue.
- Deploy the two midi buses saved from Route #46 and #45 on this route – thus operating a total of 54 one-way trips – approximately 35-minute headway.
- However, the Sankey diagram for the route (shown in the introduction) indicates possibility for even more significant optimisation. Turning radius permitting, this route can be ideally split further and operated as follows:
    - One bus along the entire route from Baramunda ISBT to Sai Mandir (10.9 km)
    - Another bus between Baramunda ISBR to Kalpana Square (7.5 km), and
    - Third bus between Fire Station Square to Sai Mandir (9.8 km)
- This optimisation will allow operating a bus every 30 minutes on the overlapping section, thus leading to significant increase in ridership.

Impact Estimate: **Ridership Increase:** 1,000 pax/day (3 midi buses together)
Revenue Increase: ₹2.65 lakh/month

ROUTE #83:

Route: *Kandarpur - Dhabaleswar*
Operating Parameters: *40.0 km | 3 Midi Bus | Non-AC | 21 One-way Trips*
Performance Indicators: ***OR:*** *29% |* ***EPKM:*** *₹16/km |* ***RL/ATL:*** *4.9*
Ridership: *407 pax/bus/day*

Figure 31: Boarding & Alighting pattern of route #83 Up



Assessment: This is an unrealistic route with an extremely high RL/ATL ratio, meaning that practically no passenger is travelling even half the length of the route. Majority of the boarding and alighting activity is happening within Cuttack city limits. The route operates at a headway of 80 to 120 minutes making it irrelevant to the urban passenger.

Recommendation: Following modifications can improve the performance of the route significantly:

- Terminate the operations between Dhabaleswar to Biju Patnaik Park (6 km) and between Kalapada Market to Kandarpur (9.7 km). This will reduce the operational route network to about 24.5 km.
- Split the route #83 into two separate but overlapping route segments operated in combination with the 3 midi buses:
  - Badambadi Bus Stand – Kalapada Market (15.1 km) operating at 60 minute headway.
  - Biju Patnaik Park – Link Road (11.5 km) operating at 30 minute headway.
  - <u>Note:</u> Planning buses in this manner may require a scheduling software. If this is to be planned manually, then the 1st segment can be operated with 1 bus at 75 minute headway, and 2 buses operated on the 2nd segment at 30 min headway.
- Further, these buses are currently operating from Patrapada Depot, resulting in very high dead mileage. These buses should be operated from the CDA Depot, thereby reducing over 50 km in dead mileage.

Impact Estimate: **Ridership Increase:** about 500 pax/day (3 midi buses together)
Revenue Increase: ₹1.60 lakh/month

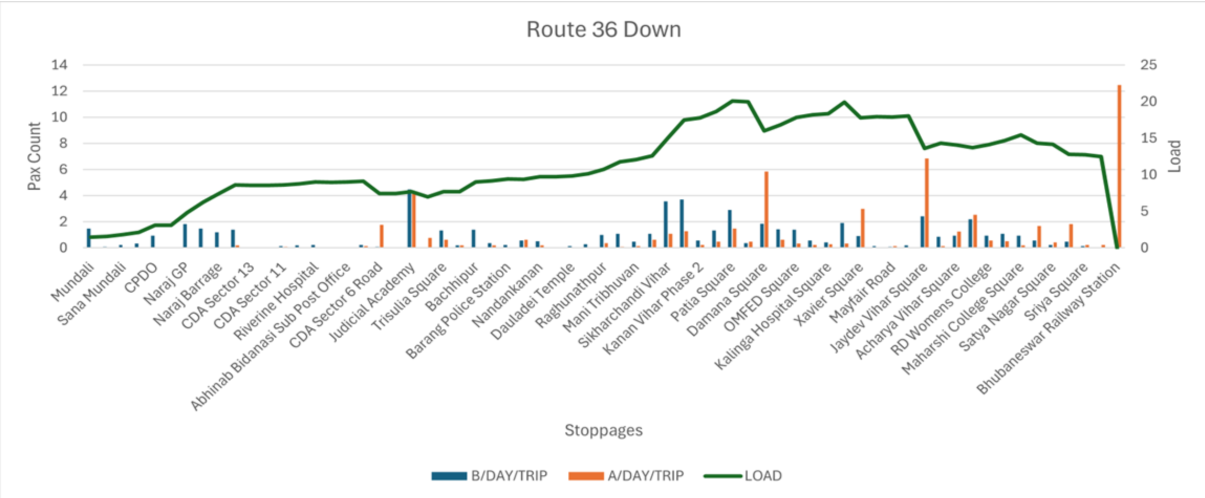
ROUTE #36:

Route: Bhubaneswar Railway Station - Mundali

Operating Parameters: 42.6 km | 1 Midi Bus | Non-AC | 4 One-way Trips

Performance Indicators: **OR:** 32% | **EPKM:** ₹18/km | **RL/ATL:** 5.6
Ridership: 341 pax/bus/day

Figure 32: Boarding & Alighting pattern of route #36 Dn



Assessment: RL/ATL of 5.6 clearly indicates a recklessly planned route on the basis of meaningless requests from vested groups. It is very much possible that there is passenger demand at Mundali. That does not translate into passenger ridership automatically unless operated at logical timings. Connecting Mundali to Bhubaneswar with just one bus was clearly meaningless. Less than 20 passengers board MoBus between Mundali and CDA Sector-11 each day.

Recommendation: This route should be terminated with immediate effect.

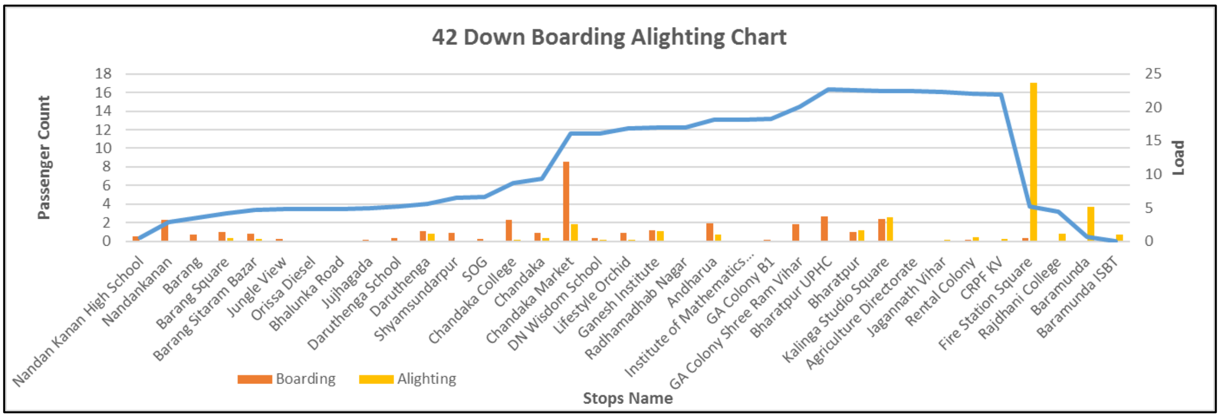
- The bus can be either used as a shuttle service between Mundali – Biju Patnaik Park (in Cuttack) – Dhabaleswar if CRUT can not avoid extending service to these areas, OR
- The bus can be deployed on Route #12, which is a reasonably high demand route with potential EPKM of ₹35/km.

Impact Estimate: **Ridership Increase:** 300 pax/day
Revenue Increase: ₹0.96 lakh/month

ROUTE #42:

Route: Baramunda ISBT - Nandan Kanan High School
Operating Parameters: 23.8 km | 2 Midi Bus | Non-AC | 20 One-way Trips
Performance Indicators: **OR:** 32% | **EPKM:** ₹18/km | **RL/ATL:** 3.1
Ridership: 420 pax/bus/day

Figure 33: Boarding & Alighting pattern on route #42Dn



Assessment: This route has very good passenger load between Fire Station Square and Chandaka Market (12.0 km), despite 90-minute headway at which the 2 buses operate. The buses however run near empty between Chandaka Market to Nanadan Kanan High School.

Recommendation: Following changes are recommended for this route:

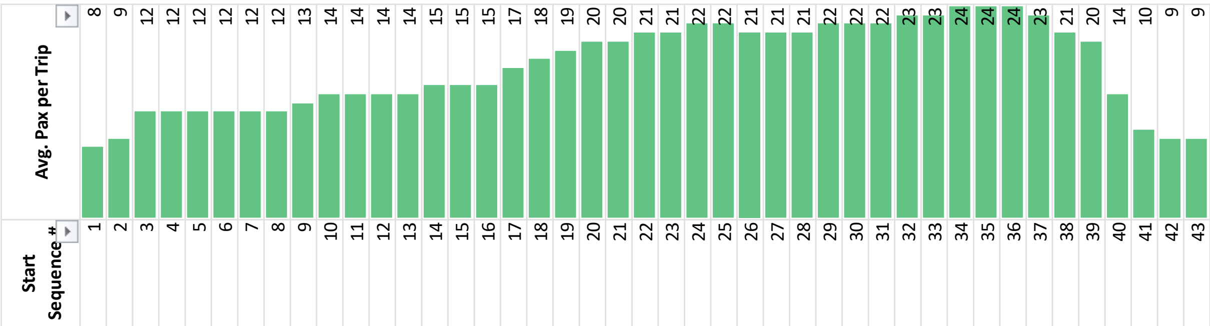
- Operate the bus only between Baramunda ISBT and Chandaka Market (13.1 km). In no case, this bus deserves to be operated beyond Chandaka College bus stop. This will allow the bus to be operated at 50 minute headway instead of the current 90 min. This will significantly improve the passenger experience and potentially increase the EPKM to ₹32/km
- One the route achieves an EPKM over ₹30/km, it is recommended that one additional midi-bus is operated along this route so that the route is operated at 20 minute headway.

Impact Estimate: **Ridership Increase:** 550 pax/day (2 midi buses together)
Revenue Increase: ₹2.1 lakh/month

ROUTE #35:

Route: Bhubaneswar Railway Station - Udayanath College
Operating Parameters: 33.1 km | 3 Midi Bus | Non-AC | 18 One-way Trips
Performance Indicators: **OR:** 37% | **EPKM:** ₹19/km | **RL/ATL:** 3.2
Ridership: 330 pax/bus/day

Figure 34: Passenger Load – Route #35 (Dn): Udayanath College - Bhubaneswar Railway Station



Assessment: On an average, the buses on this route are just half to its capacity even in the peak section. This is possibly due to the poor frequency. Also, it is possible that the ridership is more only during certain hours of the day (open and closing hours of educational institutions).

Recommendation: This route can be optimised for performance by doing the following:

- The regular route should only be operated between Bhunaneswar Rly. Stn. and Ramachandrapur stops (18.1 km) or maximum up to Prataprudrapur (26 km). This will increase the frequency of service thus making it better for the regular passengers.
- However, to not inconvenience the students who may be traveling till Udayanath College, the buses can be extended till there only for 1 hour each during the college opening and closing timings. Such caliberated scheduling will potentially need the use of software tools.

Impact Estimate: **Ridership Increase:** 200 pax/day (3 midi buses together)
Revenue Increase: ₹0.80 lakh/month

ROUTE #25:

Route: Ranasinghpur - Gadakan Yatra Padia
Operating Parameters: 26 km | 5 AC Midi & 8 Non-AC Midi | 100 One-way Trips
Performance Indicators: **OR:** 40-53% | **EPKM:** ₹37-34/km | **RL/ATL:** 5.2
Ridership: 830 pax/bus/day

Assessment: This route is a classic case of relatively high frequency, relatively high earning and very high ridership route, and yet offers lot of room for operational efficiency. Just a look at the route geography should be enough to know that it is an illogical route – meaning, no passenger going from one end of the trip (say Ranasinghpur) would follow this route to travel to the other end (Gadakan Yatra Padia). (see map & Sankey diagram in the figure below). Since this is a relatively high frequency route, many passengers traveling just 5-7 km distance take this bus. The route operates at crush loads during peak hours between Khandagiri Square till VSS Nagar Road stops but becomes near empty by the time it reaches Mancheswar Rly. Station.

Figure 35: Passengers flow analysis across route # 25

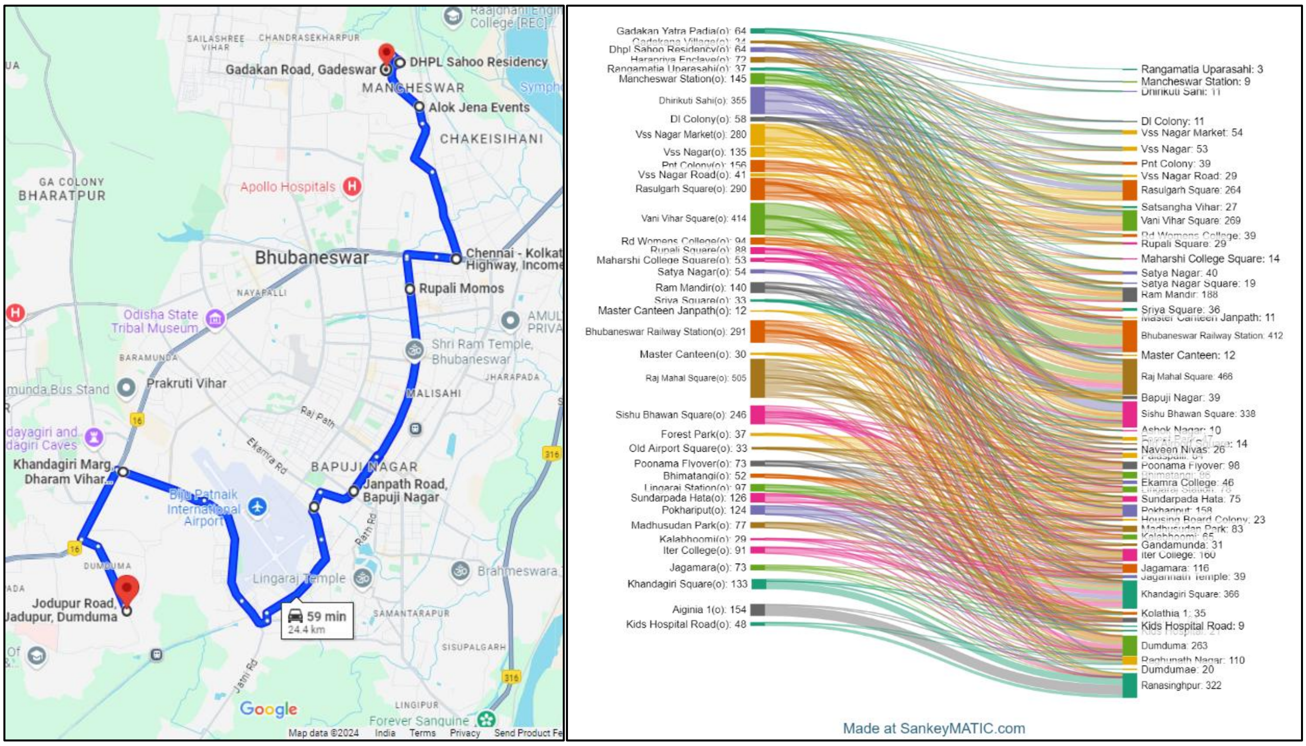
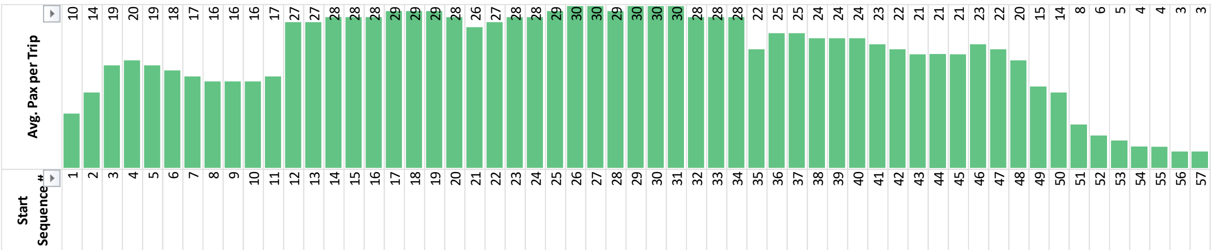


Figure 36: Passenger Load – Route #25 (Dn): Ranasinghpur - Gadakan Yatra Padia



Recommendation: This route can be optimised for performance by doing the following:

- Terminate the route at Mancheswar Rly. Stn, thereby reducing 2 km or route length that is severely underperforming. This segment is better served by auto-rickshaws.
- The remaining route can be further split into two combinations:
  - Route #25R: Ranasinghpur TO Rasulgarh Square (22.5 km) with 7 midi buses
  - Route #25M: Khandagiri Square TO Mancheswar Station (20 km) with 8 buses
- This optimisation will enable CRUT to operate 120 trips (20% more) along this stretch with the same number of buses, thereby improving the frequency for passengers and increase in ridership for CRUT.

- However, it should be noted that operating buses in this manner will require holding area for buses at Khandagiri Square, and Rasulgarh stops.

Impact Estimate: **Ridership Increase:** 1000 - 2000 pax/day (13 midi buses together)
Revenue Increase: up to 6.0 lakh/month

ROUTE #47:

Route: Settlement Office – IGKC Hospital
Operating Parameters: 39.5 km | 1 Midi Bus | Non-AC | 6 One-way Trips
Performance Indicators: **OR:** 42% | **EPKM:** ₹23/km | **RL/ATL:** 4.9
Ridership: 517 pax/bus/day

Figure 37: Passenger Load – Route #47 (Up): IGKC Hospital – Settlement Office

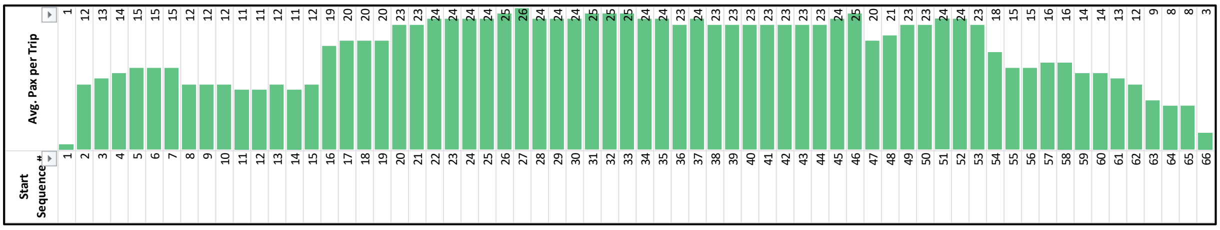
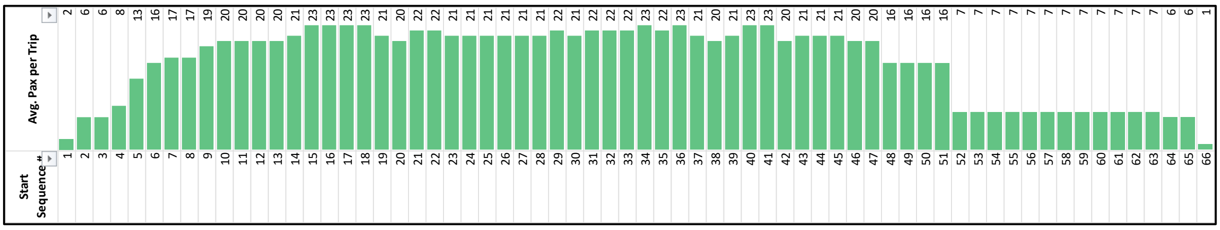


Figure 38: Passenger Load – Route #47 (Down): Settlement Office – IGKC Hospital



Assessment: This route is another classic case of poorly connecting places and with a bad frequency. This route performs reasonably well only between Jaydev Vihar Square and Cuttack Railway Station (28.6 km) where this route overlaps with Route #18. It is interesting to see reasonable ridership between Sum Hospital and Jayadev Vihar Square (7.3 km) – the place of newly developed mixed use neighbourhoods – in spite of the very poor frequency that the route offers.

Recommendation: Following changes can greatly improve the passenger experience as well as ease of planning for CRUT:

- For the major portion of the route between Jaydev Vihar Square and Cuttack Railway Station, it is suggested to add more buses to Route #18 itself. This eases the effort in scheduling for CRUT and at the same time allows for ease of communication to the passengers about increased frequency.
- Route #47 should instead be operated as a shuttle only between Sum Hospital and Jayadev Vihar Square. The shuttle service can initially operate with just this one bus which offer 1 trip/hour in each direction. Based on the performance, this short stretch

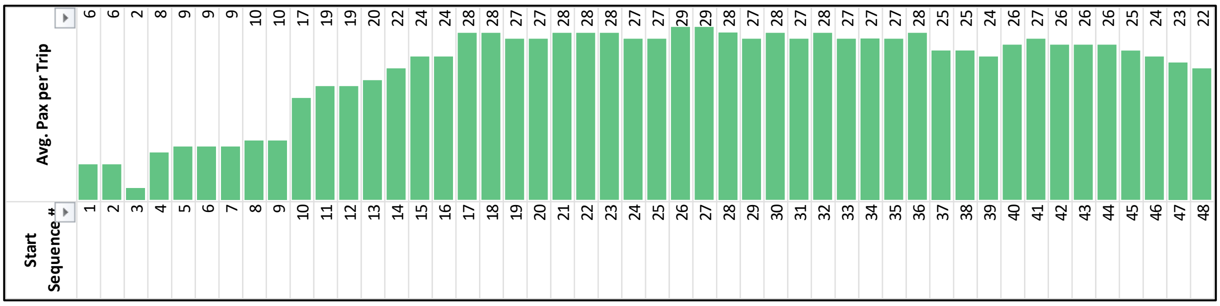
that runs parallel to the highway should either be operated at 15 minute headway or completely terminated and allow the route to be served by IPT modes.

Impact Estimate: **Ridership Increase:** 200 pax/day
Revenue Increase: ₹0.35 lakh/month

ROUTE #64:

Route: Bhubaneswar Railway Station – Jatani Gate
Operating Parameters: 32 km | 1 Midi Bus | Non-AC | 8 One-way Trips
Performance Indicators: **OR:** 45% | **EPKM:** ₹27/km | **RL/ATL:** 4.8
Ridership: 737 pax/bus/day

Figure 39: Passenger Load – Route #64 (Down): Jatani Gate – Bhubaneswar Railway Station



Assessment: This route follows the same patter as in Route #47 discussed earlier. Route #64 essentially same as route #20 which operates between Khorda and Bhubaneswar Rly. Stn. but earns 65% or the revenue of the later. The high RL/ATL is a clear indication that some portion of the route is very ineffective which can also be seen in the load graph above. The route does not provide any meaningful service to the Info Valley – Madanpur areas that this route attempts to connect. Further, the route numbering is also incorrect as the other 60's series routes are typically long distance.

Recommendation: Following changes can greatly improve the passenger experience as well as ease of planning for CRUT:

- For the major portion of the route between Jatani Gate and Bhubaneswar Railway Station, it is suggested to add more buses to Route #20 itself. This will eases the effort in scheduling for CRUT and at the same time allows for ease of communication to the passengers about increased frequency.
- It is best to operate the Info Valley – Madanpur detour as a dedicated shuttle service during the timings of the offices and educational institutions in the area. Similar to Route #47, this can first be tried with 1 midi bus but later either increase the fleet or terminate it based on the ridership and revenue earned.

Impact Estimate: **Ridership Increase:** No Change
Revenue Increase: ₹0.54 lakh/month

ROUTE #26:

Route: Jadupur – Chakeisiani

Operating Parameters: 23.9 km | 3 AC Midi & 4 Non-AC Midi | 48 One-way Trips

Performance Indicators: **OR:** 37-56% | **EPKM:** ₹35-36/km | **RL/ATL:** 4.5
Ridership: 500-740 pax/bus/day

Assessment & Recommendations: Same as route #25

8.4.2 Routes Recommended for Merger

As seen earlier, there are many routes that follow the same pattern of inefficiency, such as – single bus operations, very low frequency, illogical routes, identical routes, etc. In the interest of ease, recommendations for few routes that can be merged with the others are presented in a tabular form here:

Figure 40: List of recommended routes for merger

Route Details	Recommendations
ROUTE #41: Baramunda ISBT – DRIEMS 50 km | 2 Non-AC Midi | 12 Trips **OR:** 46% | **EPKM:** ₹24/km | **RL/ATL:** 4.4 **Ridership:** 500 pax/bus/day	Route has very poor ridership beyond Cuttack. Terminate service to DRIEMS beyond Badambadi Bus Stand. Merge this route with Route #19.
ROUTE #43: Baramunda ISBT – Banamalipur Medical 39 km | 1 Non-AC Midi | 6 Trips **OR:** 47% | **EPKM:** ₹25/km | **RL/ATL:** 4.3 **Ridership:** 500 pax/bus/day	Route has very poor ridership beyond Lingaraj Temple. Terminate service between Lingaraj Temple and Banamalipur Medical. Merge this route with Route #32.
ROUTE #44, #84, #63: Toward SVNIRTAR Olatpur 40-56 km | 3 Non-AC Midi | 18 Trips **OR:** 37-43% | **EPKM:** ₹19-20/km | **RL/ATL:** 3.7-4.3 **Ridership:** 420 pax/bus/day	Poorly planned overlapping routes that provide sparse service. No demand between Udayanath College and Madhabananda Temple – the segment needs to be terminated. Operate one extra bus along Route #16 Operate remaining two buses as shuttle service between Nakhara Square and Udayanath College (22 km) offering 32 trips in a day instead of the current 18 trips.
ROUTE #38: Bhubaneswar Railway Station Taraboi 43.5 km | 1 Non-AC Midi | 6 Trips **OR:** 48% | **EPKM:** ₹25/km | **RL/ATL:** 4.1 **Ridership:** 530 pax/bus/day	Very low frequency service which overlaps with Route #20 for most part. The deviations from Khordha Bypass Square to Taraboi (13.9 km) is used by less than 80 pax/day making the route inefficient. Merging this bus with Route #20 will increase EPKM to ₹36/km earning additional ₹0.85 lakh/mo.
ROUTE #11, #13: Toward Nandankanan via Infocity 21-27 km | 21 AC & Non-AC | 170 Trips	Both are significantly overlapping routes providing frequent service (~10 min headway) to the passengers thereby attracting high

OR: 45-50% | **EPKM:** ₹49-51/km | **RL/ATL:** 3.4-4.2 **Ridership:** 800 pax/bus/day	ridership. However, the low occupancy is a result of the segment between KIIT Square to Nandankanan being underutilized due to the detour via Infocity. By terminating both the routes at KIIT Square, these buses can operate 25% more trips, thereby improving the earning much more. To compensate for the reduced service from Nandankanan to Bhubaneswar Rly. Stn., additional buses can be operated on Route #12 which is also a high earning route.

8.4.3 Recommendations for Swapping of Buses

The section on 'Comparison of Non-AC and AC Bus Ridership' shows few routes where AC bus revenue is lower than that of the non-AC buses – meaning, if CRUT only collects non-AC bus fares on these buses, then more passengers would board the buses and thus resulting in higher revenue. This situation requires careful consideration as collecting non-AC fares on AC buses on certain routes may not be an easy option both technologically or in terms of communicating to the passengers. Therefore, below adjustments to the bus fleet are proposed to optimise the use of fleet.

ROUTE #20: Bhubaneswar Railway Station – Khordha New Bus Stand

This route is currently operated by 4 Standard AC buses and 6 Midi non-AC buses. The occupancy ration in the standard size AC buses is mere 45%. It might be best to replace the 4 standard AC buses with 4 midi AC buses thereby reducing cost of operations and using the bus capacity more efficiently.

ROUTE #12: Bhubaneswar Railway Station – Nandankanan

Same as route #20. The occupancy ration in the standard size AC buses is 34% only. Replacing the 2 standard AC buses with 2 midi AC buses thereby reducing cost of operations and using the bus capacity more efficiently.

ROUTE #70: Bhubaneswar Railway Station – Konark

The AC bus EPKM on this route is 11% lower than that earned in the non-AC bus indicating overwhelming passenger preference for non-AC bus fares. Therefore, it may be best to replace the 2 standard AC buses with 2 standard non-AC buses, thereby reducing cost of operations by 30% and increasing revenue by 11%, effectively reducing the financial viability gap by ₹3.57 lakhs/month.

ROUTE #50: Bhubaneswar Railway Station – Puri Bus Stand

On this route the 2 Standard AC buses earn 4% lower revenue than the non-AC buses. Currently, both the AC buses operate from Patia Depot and the two non-AC buses operate from Puri Depot. It may be worthwhile to interchange one AC bus from Patia with the non-AC bus from Puri to check if this improves the revenue of the AC bus.

ROUTE #26: Jadupur – Chakeisiani

This route is operated by 3 AC Midi & 4 Non-AC Midi buses. The EPKM of AC buses is 3% lesser than the non-AC fleet. The route changes suggested earlier may potentially also improve the performance of the AC buses. However, if that does not happen, then it may be worthwhile to replace the AC buses with non-AC midi buses

GUIDELINES FOR ADDITIONAL FLEET DEPLOYMENT

9. Standard Operating Procedure for Route Changes

The assessment in the previous section highlights that CRUT's route planning is currently not a technical exercise, but rather is driven by requests from interest groups without sufficient pre-implementation analysis or regular post-implementation review. This approach can adversely affect passengers, as CRUT is unable to scale its fleet as quickly as it expands its route network. To address these issues, it is crucial for CRUT to establish a Standard Operating Procedure (SOP) for making route changes, which should include the following guidelines:

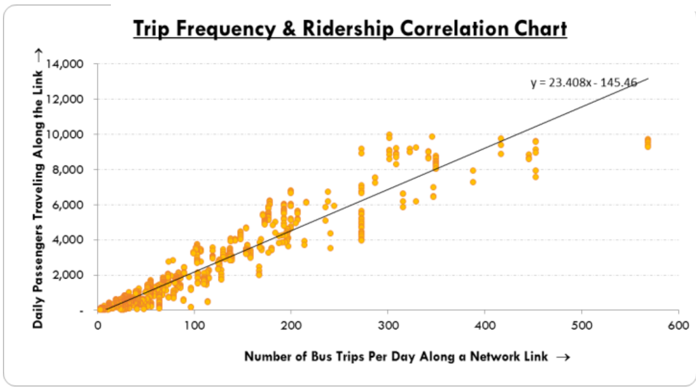
- **Scheduled Implementation:** Route changes should be implemented only on the first day of each quarter.
- **Request Submission Timeline:** Only requests received within the first month of the current quarter will be eligible for consideration and possible implementation in the following quarter. Requests received after this period will be evaluated in the subsequent quarter. This process means that the minimum time required for a route change, from request to implementation, is two months, but it can take up to five months.
- **Route Examination Criteria:** Any new route proposal must undergo a thorough assessment, including the estimation of potential demand, the required level of service, and the type of bus fleet to be deployed. A route change should be approved only if all of the following conditions are met:
    - The route has sufficient demand to support a bus operating at a 30-minute headway with at least 50% occupancy. If this condition is not met, an autorickshaw service may be more appropriate.
    - CRUT has adequate fleet capacity to operate the new route or extend an existing route at the required headway.
    - If fleet resources are reallocated from another route, the key performance indicators (KPIs)—such as Occupancy Rate (OR), Earnings Per Kilometre (EPKM), and Fleet Utilization—must be benchmarked, and the new route should enhance these KPIs.
- **Preparation and Digitization:** The route changes should be examined and approved in the second month of the quarter. Thereafter, the new stops, routes, and fare tables should be finalized in the same month. These changes must be digitized in the ITS system and updated in the mobile app in the 1st week of the last month in the quarter.

- **Communication of Changes:** Route changes should be communicated to the public through information posters in relevant bus routes, through various media channels, and press releases during the second and fourth weeks of the third month. The changes will then be implemented on the first day of the next quarter.
- **Performance Monitoring:** The revised routes should be continuously monitored on a weekly basis to evaluate performance. Adjustments can be made weekly based on feedback. Key considerations include:
    - Ensuring that the ratio of Route Length (RL) to Average Passenger Trip Length (ATL) does not exceed 3.3.
    - If the route fails to meet the benchmarked KPIs, the route will be discontinued after a one-month trial period, and the buses will be redeployed to their original routes.

9.1 Guidance for New Routes

- **FREQUENCY IS FREEDOM:** Urban bus transport demand is directly proportional to the frequency of service. The adjacent graph illustrates CRUT's bus route network links plotted against the frequency of service (number of bus trips per day) and ridership along each route segment. It is evident that most data points for network links serviced by fewer than 100 bus trips per day fall below the trend line, indicating a lower passenger-to-trip ratio. In contrast, network links with 100 to 350 trips per day generally lie above the trend line, reflecting a higher ratio of passengers per trip.

Figure 41: Trip frequency & Ridership correlation chart



Note: Links having greater than 400 trips also fall below the trend line. This is because these are network links that are next to passenger terminals – Bhubaneswar Rly. Stn, Rajmahal Square, Baramunda ISBT, etc. Thought this may be unavoidable due to lack of additional turning points or holding areas for the buses, the lower ratio indicates that some of the bus trips can be shifted out of these terminals.

The route rationalization exercise should prioritize network links that fall below the trend line—those with a lower passenger ridership compared to the number of trips operated. These links should be evaluated to determine whether to reduce, increase, or eliminate transit service to shift these points above the trend line. A sample of network links for targeted action is presented in the following section.

Assuming bus trips are uniformly distributed over 16 hours of daily operation, 350 bus trips equate to a bus every 2-3 minutes, which is the ideal frequency for urban transit operations. However, because not all buses are headed to the same destination or direction, such a frequency may still result in passenger wait times of up to 10 minutes if there are too many routes. Therefore, transit planners should also limit the number of routes to optimize efficiency.

- **NETWORK TO FLEET RATIO:**
  Building on the findings from the previous principle, there is a thumb rule for the ratio of route network length to fleet size in urban operations. Ideally, this ratio should be between 0.5 and 1.0, meaning there should be at least one bus for every operational kilometre (preferably two buses per kilometre) of route network. Operating a dense transit network provides several advantages:
  - Ensures highly reliable service for passengers.
  - Increases transit mode share, reducing traffic congestion.
  - Simplifies operations, including route planning, monitoring, and infrastructure management, reducing bus stops, terminals, and ticket-checking resources.
  - Minimizes the financial viability gap within the existing fare structure.

  Currently, CRUT operates about 900 km of route network with a fleet of only 300 buses, which is three times more than the optimal ratio. This imbalance poses significant challenges in terms of sustainability and passenger satisfaction.

- **MAXIMUM HEADWAY**
  Given that urban passenger trips are typically short—averaging 8.8 km per trip for MoBus—passengers generally prefer not to wait long for short journeys. CRUT should aim to maintain a maximum headway of 10 minutes to minimize passenger waiting time and enhance the service experience.

- **RATIO OF ROUTE LENGTH TO AVERAGE PASSENGER TRIP LENGTH (RL/ATL)**
  This ratio is a key indicator of route planning efficiency relative to passenger travel demand. The ideal range for the RL/ATL ratio is between 1.8 and 2.6, but it should never exceed 3.0. An RL/ATL greater than 3.0 clearly indicates a need for route optimization. Currently, 42 of the 65 bus routes operated by CRUT have RL/ATL greater than 3.0.

- **ROUTE NOMENCLATURE**
  CRUT currently operates overwhelming number of routes for the small bus fleet that it operates. While the changes suggested in this document will reduce a few bus routes, it is concerning that CRUT assigns route numbers randomly making it very difficult for passengers to decipher the direction/destination to which the route is operating to. Therefore it is highly recommended that CRUT comes up with a nomenclature schema using both numbers and alphabets, and in a gradual manner switches the route numbers to the new numbering scheme.

Together, these four principles help maintain an optimal transit network given the available fleet size. It is important to note that the route changes recommended in this document aim to reduce the transit service network by about 20%. CRUT should plan to deploy its additional fleet along the optimized network until it meets the four operational parameters outlined here.

9.2 Categorising Route Segments by Performance

		Ratio: (Daily Pax Load)/(Daily Trips)						Total
		0 - 7	8 - 11	12 - 15	16 - 19	20 - 24	>= 25	
Daily Trips	3 - 4	158	124	79	57	31	3	452
	5 - 11	74	92	50	12	4	1	233
	12 - 25	36	72	90	69	61	26	354
	26 - 40	32	68	67	101	53	21	342
	40 - 100	50	29	39	62	128	42	350
	100 - 400	2		21	42	104	182	351
	> 400				10	8		18
Total		352	385	346	353	389	275	2100

Colour	Meaning
Green	⇒ Links with relatively high ridership and may benefit from increase in trips
Red	⇒ Network links that are underperforming. Can examine service rationalisation
Yellow	⇒ Segments need re-evaluation. Either increase trips or eliminate from network

The scatter plot of transit network segments shown in the previous section (figure # 41)can be cross categorised in a matrix form by the number of bus trips on the segment and passenger load per trip – shown here. This representation helps in easily identifying route links for further action as colour coded in the matrix. The lighter shades represent the borderline conditions.

Thus, among the 2100 route segments in MoBus transit network, 493 segments – totalling about 265 km – are relatively having high demand and may need more bus trips. Similarly, services can clearly be reduced on 180 route segments that together constitute about 55 km of route network. Another 175 km or route network (356 links) are operating at very low frequency and are not attracting ridership. These segments need a serious evaluation to either completely eliminate services to focus on high demand areas, or number of bus trips have to be increased, to be able to provide a meaningful service the passengers can opt.

Another 276 and 157 segments are marked as borderline scenarios to test with increase and decrease in trips respectively. These route segments are spatially presented in GIS maps above. The total list of segments is presented separately in Annexure – B.

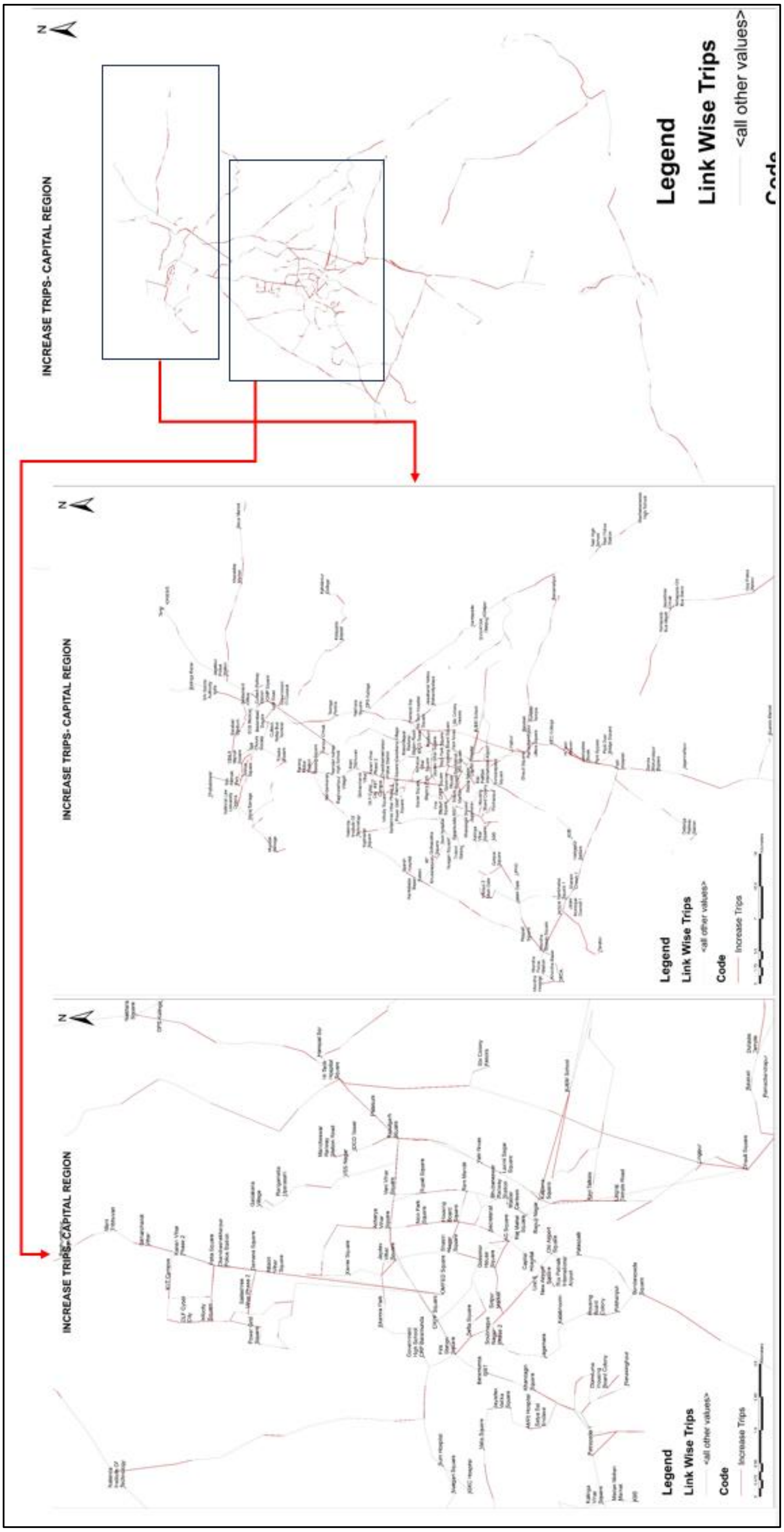


Figure 42: Graphical Representation of Network Links Where Bus Trips Can Be Increased

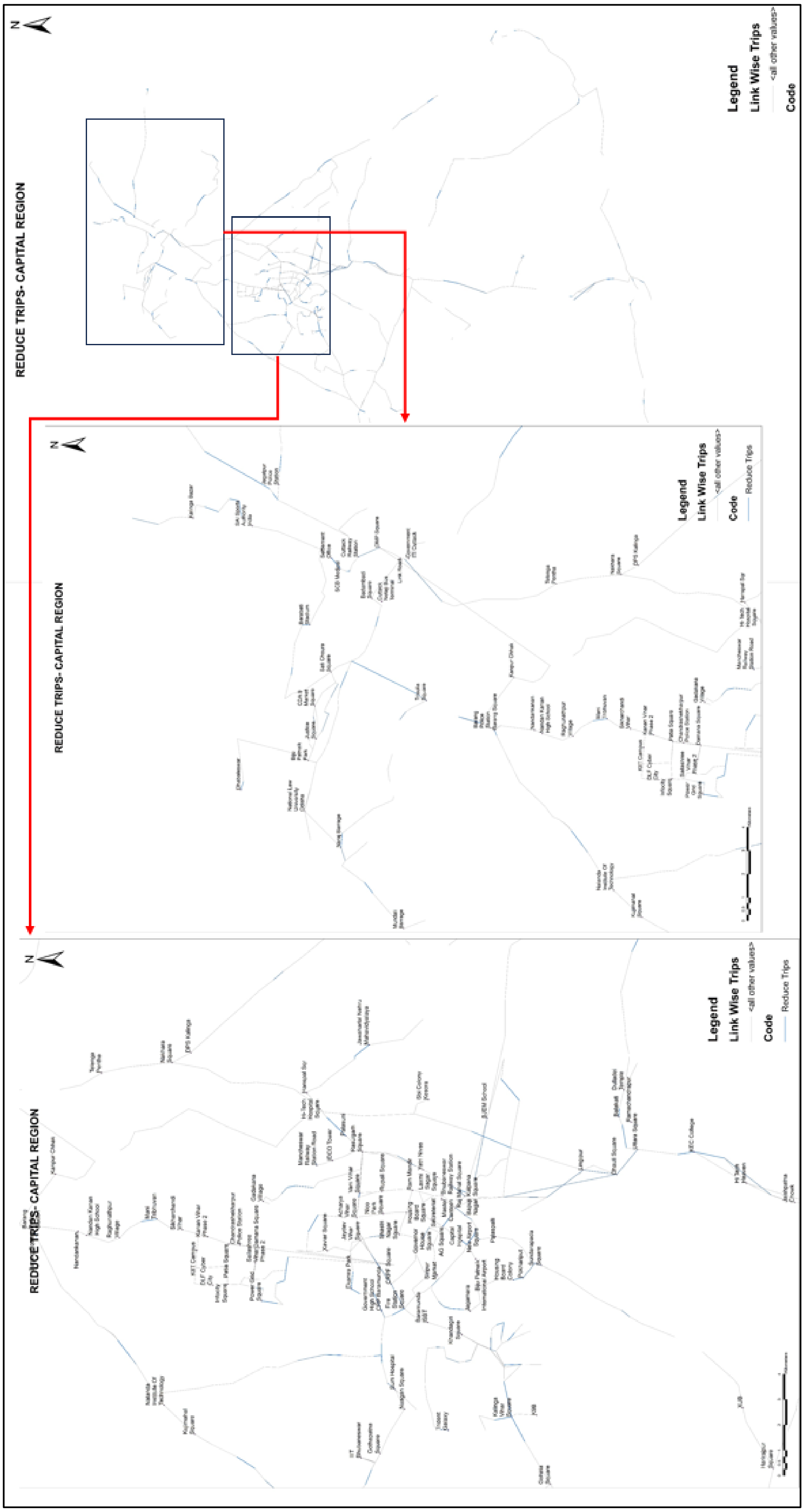


Figure 43: Graphical Representation of Network Links Where Bus Trips Can Be Reduced

10. Challenges Hindering Excellence At CRUT

CRUT faces multiple challenges in managing sustainable urban transit which need immediate attention.

1. **Data Issues from ITS**: Inaccurate route coding and data discrepancies from Intelligent Transportation Systems (ITS) complicate analysis. GPS data collection intervals are too long, causing inaccuracies in assessing bus utilization and travel times. Ideally, the ITS system should automatically calculate the KPIs, present visualizations, and flag underperforming routes so that CRUT's operations staff can focus on problem identification and decision-making. However, the inconsistencies in the data and lack of correlatability have made this analysis a tedious and time-consuming task. This hinders CRUT’s ability to optimize services effectively.

2. **Manpower Retention and Hiring**: Inconsistent hiring practices and salary structures make it challenging to attract and retain skilled employees. CRUT has been extremely customer-focused in its first six years, addressing every passenger complaint, but it has not been able to establish Standard Operating Procedures (SOPs) to reduce the ad-hoc nature of issue resolution. The lack of SOPs also hinders staff skilling and capacity building. These issues, combined with challenges in hiring and retaining skilled manpower and a non-functional ITS system, plague CRUT's ability to scale into a high-performance transit entity.

3. **External Pressures**: CRUT is often influenced by external demands from the government and local communities, which can contradict data-driven decisions. These pressures lead to maintaining underperforming routes, diverting resources from more efficient operations. Effective implementation of the SOP suggested in the earlier section can to an extent reduce such influences. At least 20 of the 65 bus routes of CRUT are being operated with a single bus with each one of them is operating sub-optimally. At least 10 other routes deserve to be consolidated by truncating the end sections of the route that are better served by auto-rickshaws.

4. **Lack of Digital Interface for Fleet Scheduling**: The absence of a digital scheduling tool limits CRUT’s ability to optimize routes and schedules. Current planning relies on public input rather than data-driven analysis, leading to inefficiencies. A functional and effective bus scheduling solution could help CRUT address frequent route change requests from interest groups by scheduling just a pair of trips at the most appropriate hours instead of dedicating an entire bus that only serves a destination once every couple of hours. This would allow CRUT to more efficiently use its fleet on high-density routes, improving passenger experience.

5. **Electric Bus Deployment**: The deployment and electrification of buses present substantial challenges. The limited range of electric buses, inadequate charging infrastructure, and unreliable power supply reduce operational efficiency, resulting in increased dead kilometers, operational delays, and a reliance on both electric and diesel buses, complicating transit schedules. Replacing diesel buses with electric ones involves substantial costs and logistical challenges, as more electric buses are needed to replace diesel ones due to different performance metrics. Strategic planning and investment are crucial to successfully electrify the fleet and realize the benefits of sustainable urban transit.

6. **Aging Bus Fleet**: The fleet is aging, with buses approaching obsolescence. Replacing them will require careful planning and significant investment to ensure uninterrupted service.

7. **Increasing Traffic Congestion**: Rising traffic congestion slows down buses, reducing service reliability and attractiveness, negatively impacting passenger experience and public transport utilization.

Addressing these challenges requires strategic planning and investment to enhance CRUT's operational efficiency and service quality.

SOFTWARE-BASED SOLUTION FOR SCHEDULING & OPERATIONAL PLANNING

11. Introduction

Bus scheduling is a complex and highly responsible function that lies at the core of efficient and reliable public transport operations. It requires a careful balance between operational efficiency, cost-effectiveness, and service quality, while addressing the needs of passengers, drivers, and management. Schedulers are responsible for determining the number of vehicles required, designing crew duties, and ensuring that services operate at appropriate frequencies across different times of the day. This process demands a high level of technical skill, operational understanding, and contextual awareness.

Effective scheduling must account for realistic travel times, adequate driver breaks, and variability in traffic conditions, while maintaining consistent and predictable timetables to enhance commuter satisfaction. Poor scheduling can lead to delays, inefficiencies, increased operational costs, and reduced reliability of services.

12. Objectives of study

The study aims to strengthen operational efficiency and service reliability of the bus network through a structured, data-driven scheduling approach. The key objectives are:

- To identify critical **operational constraints** and define relevant **Key Performance Indicators (KPIs)** that influence scheduling efficiency, service quality, and resource utilisation.
- To develop optimised **bus schedules for a fleet of 200 e-buses**, incorporating spatio-temporal demand patterns, route characteristics, peak and off-peak variations, and operational feasibility.
- To assess and document the outcomes of the scheduling exercise through a comprehensive **report and presentation**, highlighting performance improvements, operational insights, and recommendations for scalable implementation.